PROJECT TITLE : AWS VPC PEERING

NAME : RAGHAVENDRAN G M

AICTE STUDENT ID :
STUc171e6fe139bf1789379999

INTERNSHIP : AWS CLOUD COMPUTING INTERN

BATCH : 1

MODE : VIRTUAL

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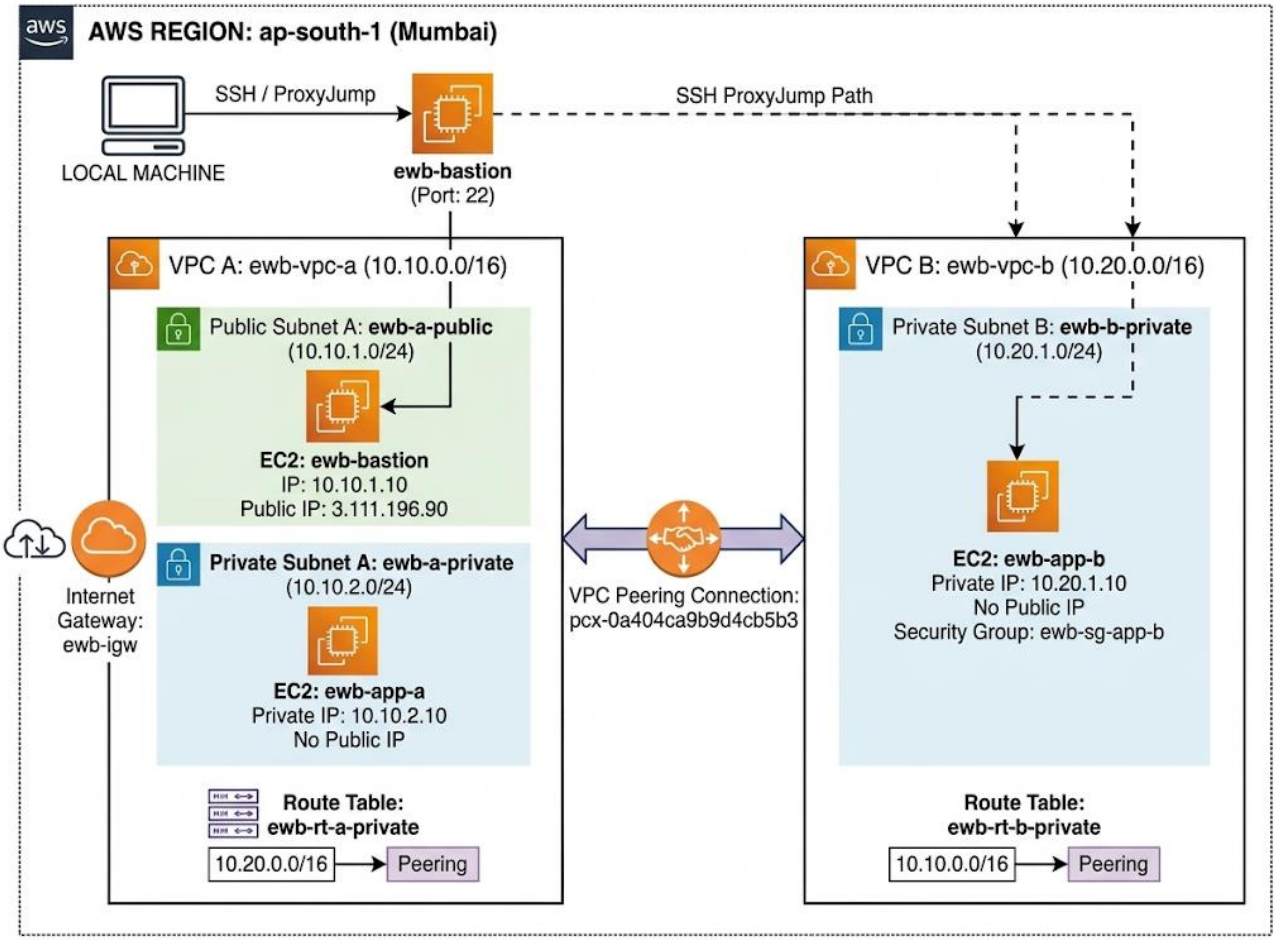
AWS REGION : MUMBAI(ap-south-1)

COMPLETION DATE : 24.10.2026

Executive Summary

This lab demonstrates the implementation, stateful security validation, and decommissioning of an intra-region AWS VPC Peering infrastructure in ap-south-1 (Mumbai). The architecture establishes private cross-VPC communication between two isolated Virtual Private Clouds without traversing the public internet, managed via local SSH ProxyJump and validated using controlled failure testing.

Architecture Diagram



Key Architectural Elements:

- **VPC A (10.10.0.0/16):** Contains a public subnet for administration and a private subnet for internal workloads.
- **VPC B (10.20.0.0/16):** Contains an isolated private subnet representing a separate application tier or workload.
- **Internet Gateway (ewb-igw**): Attached to VPC A to provide inbound/outbound internet access specifically for the public Bastion host.

- **Public Bastion Host (ewb-bastion):** Located in Public Subnet A (10.10.1.0/24) with a public IP (3.111.196.90), serving as the single entry point for administrator SSH access.
- **Private Application Servers:**
    - **ewb-app-a (10.10.2.10):** Deployed in Private Subnet A, isolated from direct internet access**.**
    - **ewb-app-b (10.20.1.10):** Deployed in Private Subnet B in VPC B, isolated from direct internet access.
- **VPC Peering Connection (pcx-0a404ca9b9d4cb5b3):** Routes private IPv4 traffic directly between VPC A and VPC B using AWS internal infrastructure.

Infrastructure Architecture & Configuration Parameters

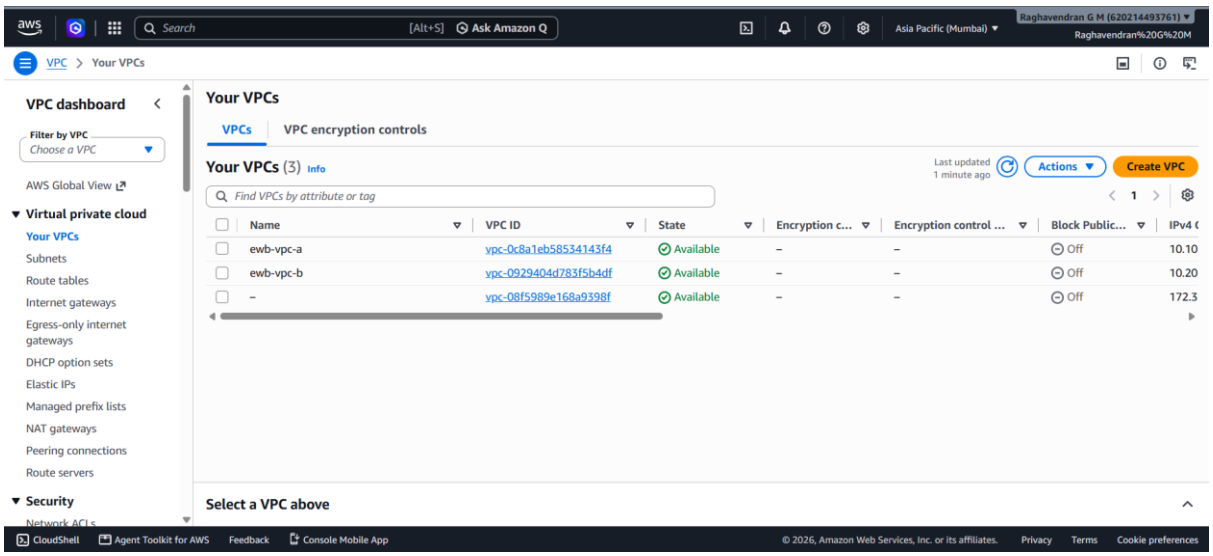
Virtual Private Clouds (VPCs) & Subnets

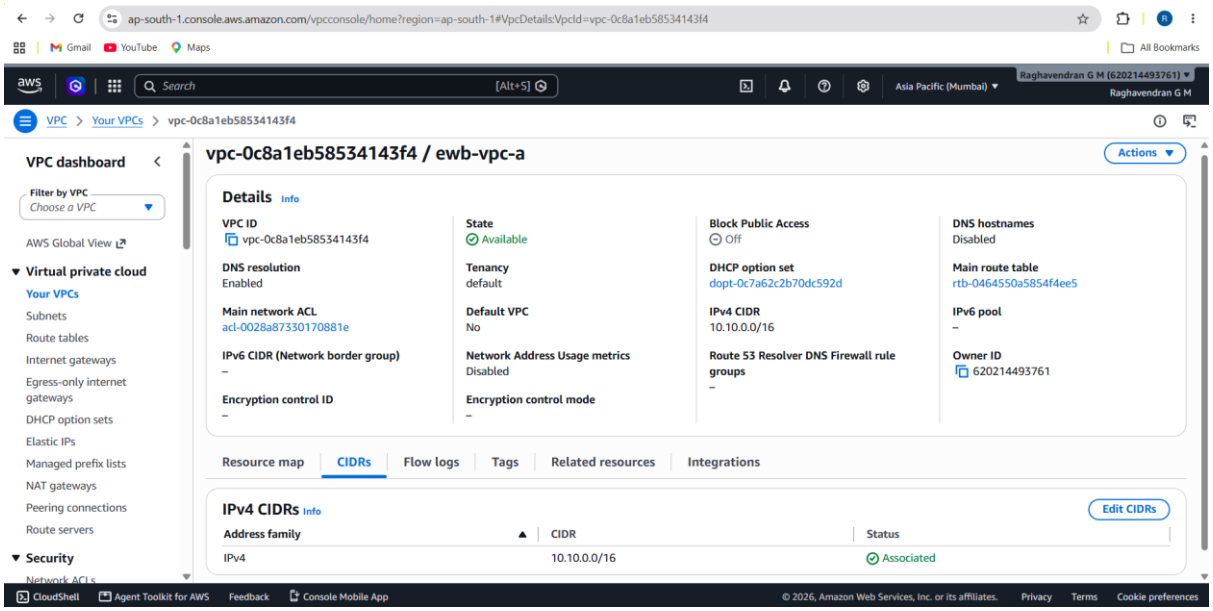
VPC A (ewb-vpc-a): vpc-0c8a1eb58534143f4 | CIDR: 10.10.0.0/16

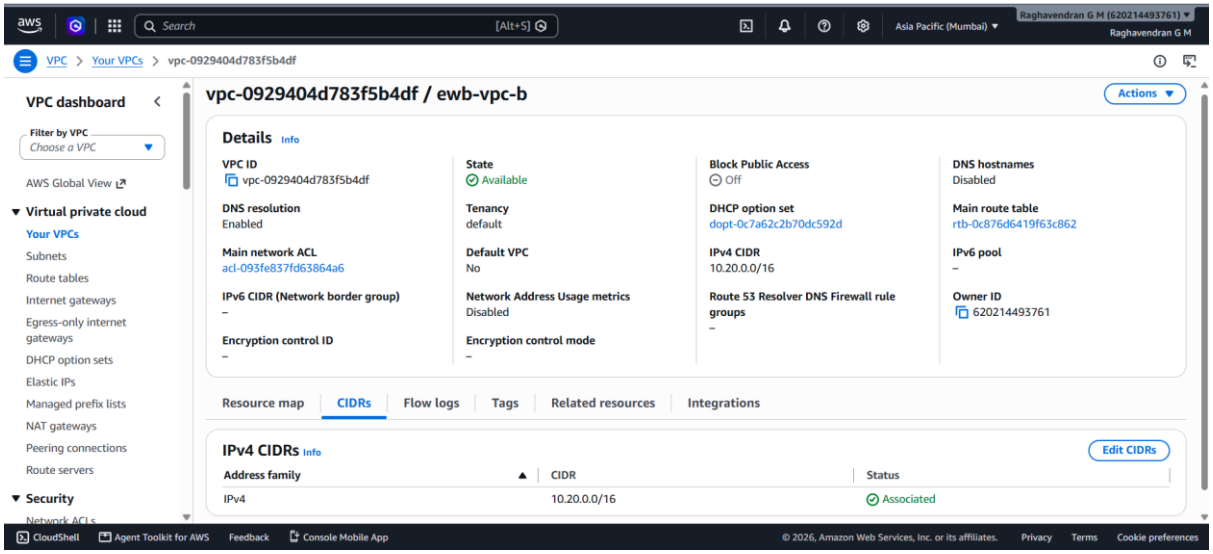
- **Public Subnet (ewb-a-public):** subnet-0c8b6afd632a13049 | CIDR: 10.10.1.0/24 | AZ: ap- south-1a
- **Private Subnet (ewb-a-private):** subnet-00920c6b4f6f7272e | CIDR: 10.10.2.0/24 | AZ: ap- south-1a

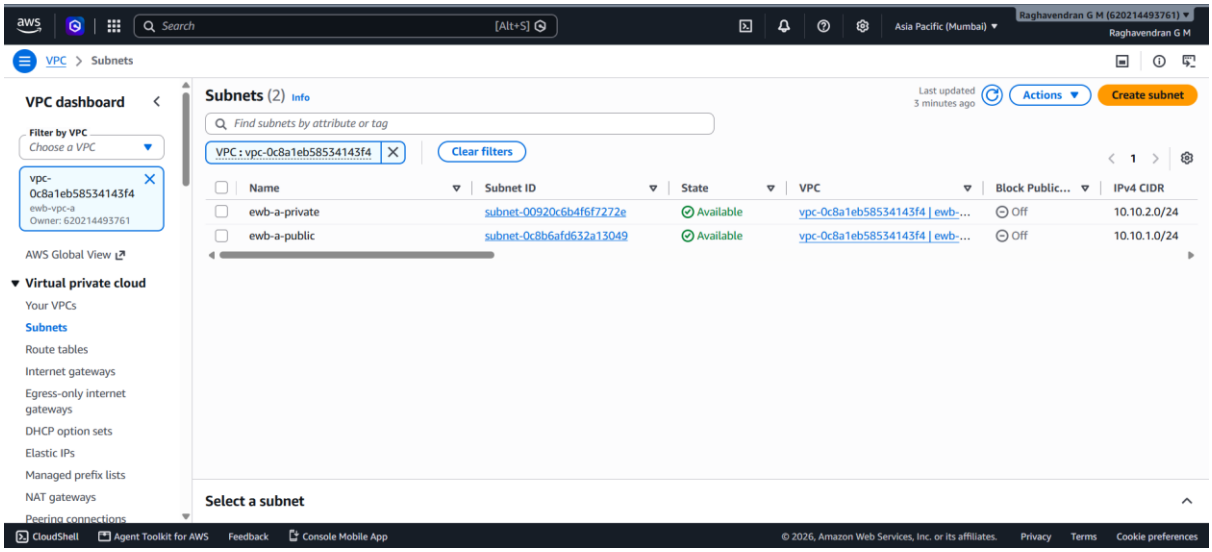
VPC B (ewb-vpc-b): vpc-0929404d783f5b4df | CIDR: 10.20.0.0/16

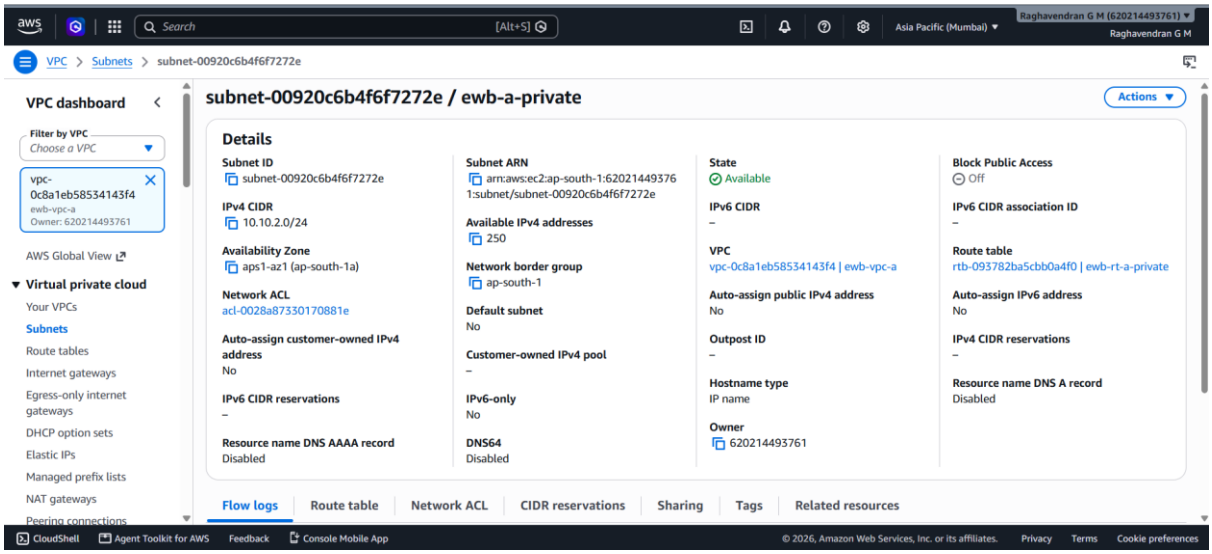
- **Private Subnet (ewb-b-private):** subnet-0df374ef519f84b01 | CIDR: 10.20.1.0/24 | AZ: ap-south-1a

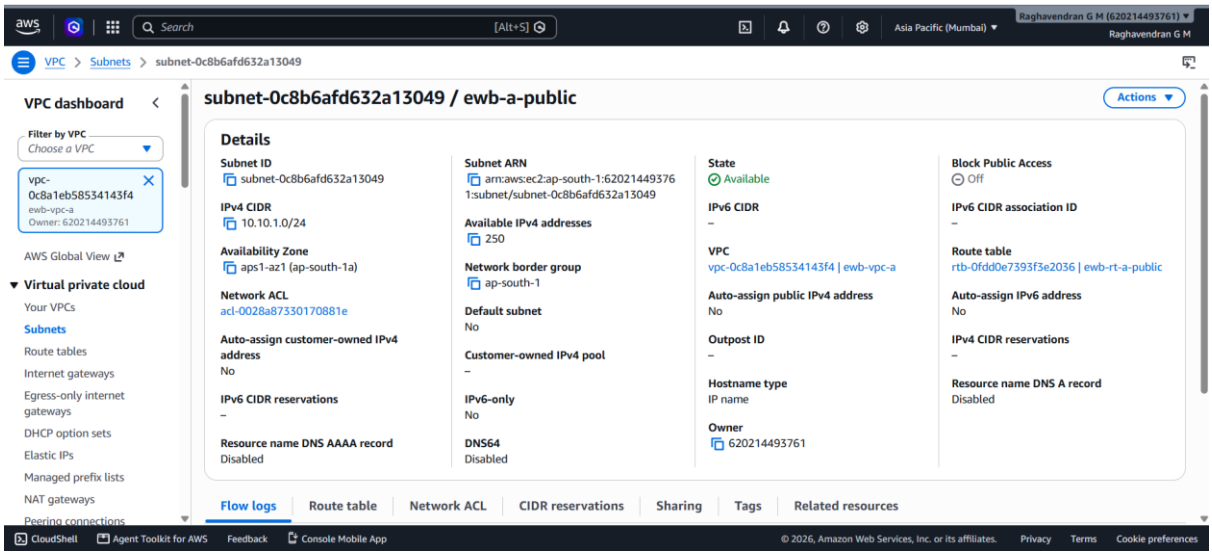


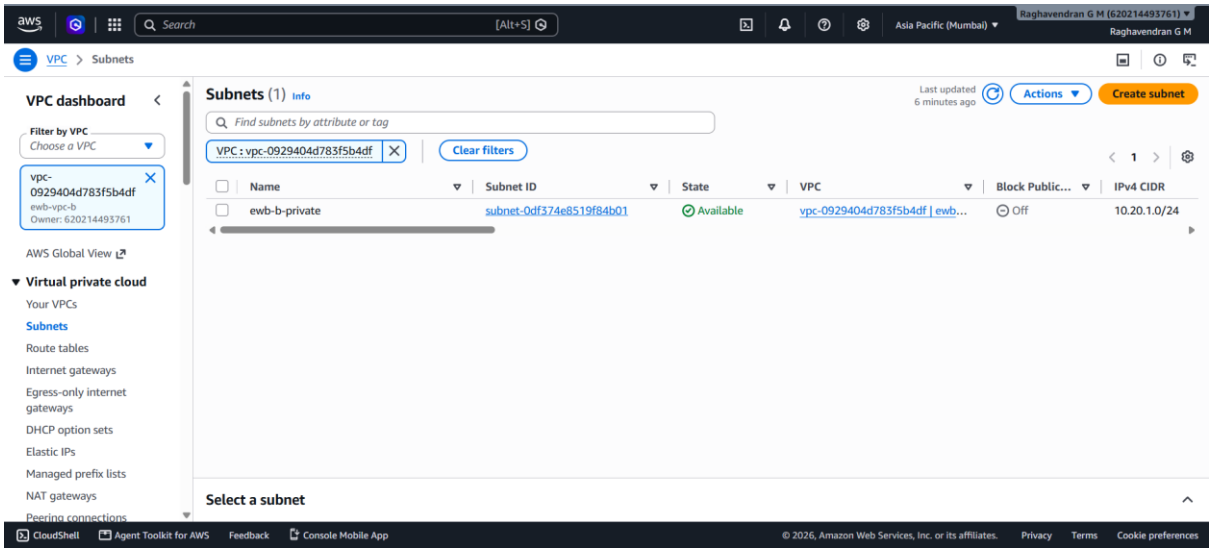

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#VpcDetails:VpcId=vpc-0c8a1eb58534143f4
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Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
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Main route table
rtb-0464550a5854f4ee5
Main network ACL
acl-0028a87330170881e
Default VPC
No
IPv4 CIDR
10.10.0.0/16
IPv6 pool
–
IPv6 CIDR (Network border group)
–
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
–
Owner ID
620214493761
Encryption control ID
–
Encryption control mode
–
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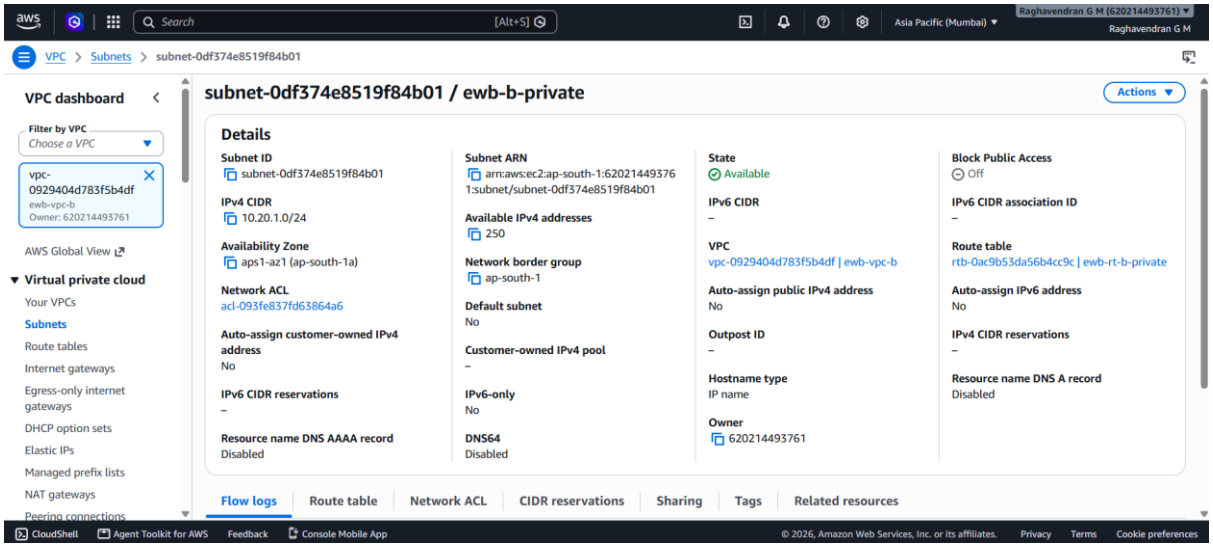

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DHCP option set
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Main route table
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Main network ACL
acl-093fe837fd63864a6
Default VPC
No
IPv4 CIDR
10.20.0.0/16
IPv6 pool
–
IPv6 CIDR (Network border group)
–
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
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Find subnets by attribute or tag
VPC : vpc-0c8a1eb58534143f4
Clear filters
Name
Subnet ID
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Block Public...
IPv4 CIDR
ewb-a-private
subnet-00920c6b4f6f7272e
Available
vpc-0c8a1eb58534143f4 | ewb-...
Off
10.10.2.0/24
ewb-a-public
subnet-0c8b6afd632a13049
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Details
Subnet ID
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IPv4 CIDR
10.10.2.0/24
Availability Zone
aps1-az1 (ap-south-1a)
Network ACL
acl-0028a87330170881e
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Resource name DNS AAAA record
Disabled
Subnet ARN
arn:aws:ec2:ap-south-1:620214493761:subnet/subnet-00920c6b4f6f7272e
Available IPv4 addresses
250
Network border group
ap-south-1
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
DNS64
Disabled
State
Available
IPv6 CIDR
–
VPC
vpc-0c8a1eb58534143f4 | ewb-vpc-a
Auto-assign public IPv4 address
No
Outpost ID
–
Hostname type
IP name
Owner
620214493761
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-093782ba5cbb0a4f0 | ewb-rt-a-private
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
Flow logs
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CIDR reservations
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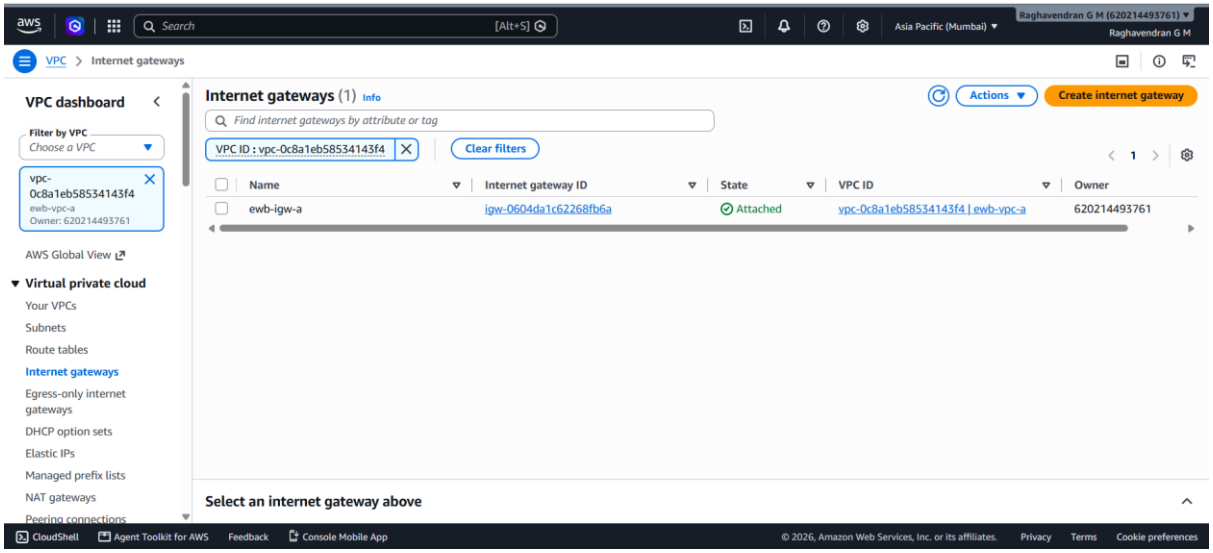

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Subnet ID
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10.10.1.0/24
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Network ACL
acl-0028a87330170881e
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Resource name DNS AAAA record
Disabled
Subnet ARN
arn:aws:ec2:ap-south-1:620214493761:subnet/subnet-0c8b6afd632a13049
Available IPv4 addresses
250
Network border group
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Default subnet
No
Customer-owned IPv4 pool
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IPv6-only
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DNS64
Disabled
State
Available
IPv6 CIDR
–
VPC
vpc-0c8a1eb58534143f4 | ewb-vpc-a
Auto-assign public IPv4 address
No
Outpost ID
–
Hostname type
IP name
Owner
620214493761
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-0fdd0e7393f3e2036 | ewb-rt-a-public
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
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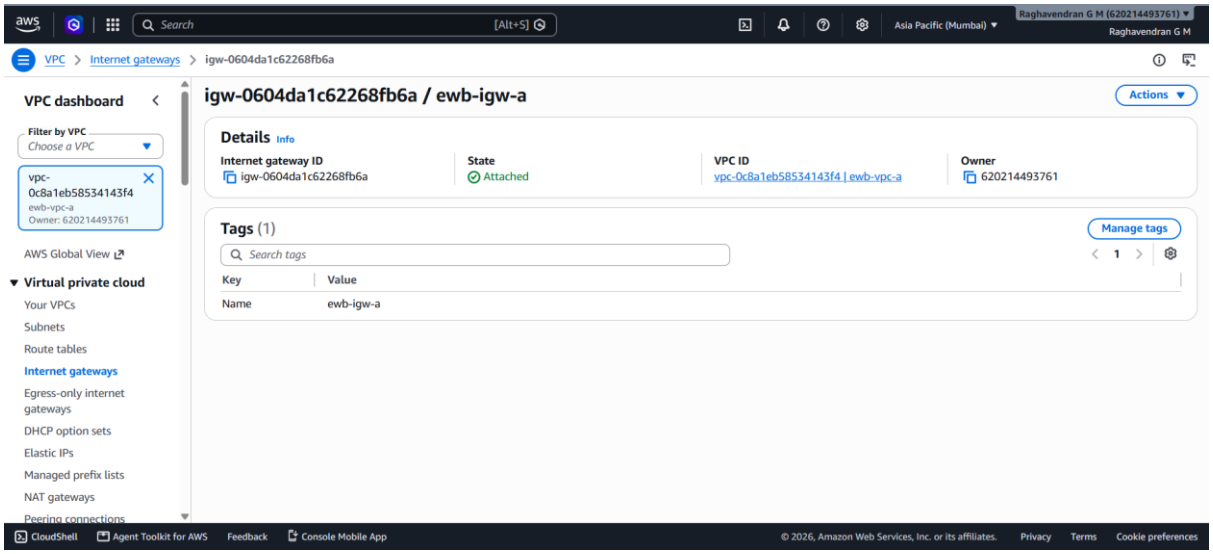




Internet Gateway & Peering Connection

- **Internet Gateway (ewb-igw-a):** igw-0604da1c62268fb6a (Attached to ewb-vpc-a)





VPC Peering and Routing

To enable communication between isolated VPCs, a VPC Peering Connection (pcx-0a404ca9b9d4cb5b3) was established between Requester ewb-vpc-a and Accepter ewb-vpc-b.

VPC Peering Connection (ewb-a-b-peer): pcx-0a404ca9b9d4cb5b3 (Requester: ewb-vpc-a | Accepter: ewb-vpc-b | Status: Active)

VPC Peering Details

Parameter	Configuration Value
Peering Connection ID	pcx-0a404ca9b9d4cb5b3
Status	Active
Requester VPC ID	vpc-0c8a1eb58534143f4 (ewb-vpc-a)
Accepter VPC ID	vpc-0929404d783f5b4df (ewb-vpc-b)
Requester CIDR	10.10.0.0/16
Accepter CIDR	10.20.0.0/16

Route Table Configuration

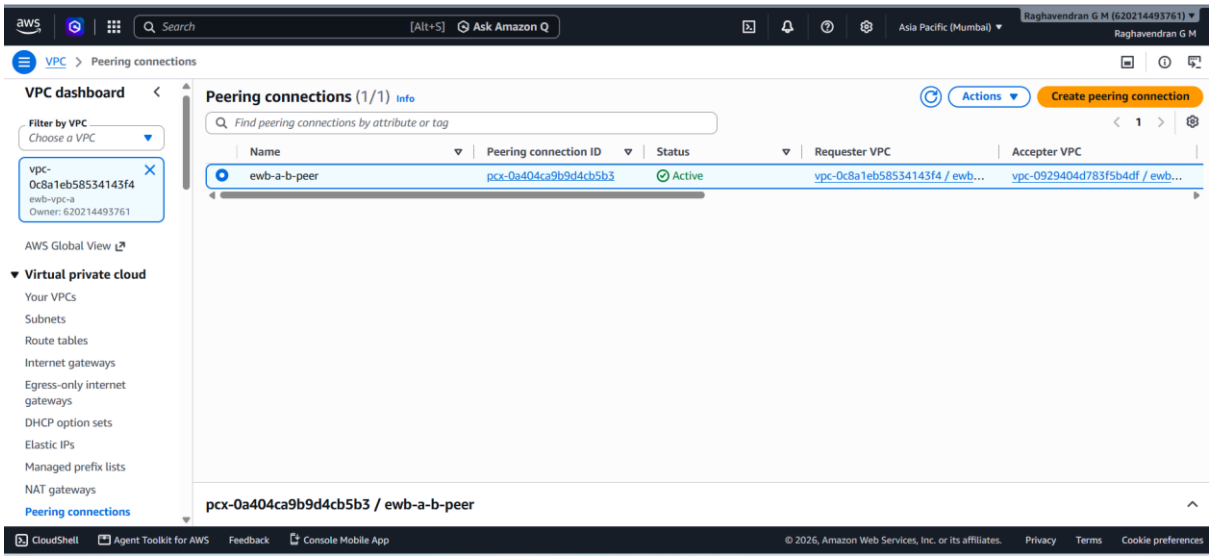
To direct cross-VPC traffic across the peering connection, specific routes were added to the respective private subnet route tables in both VPCs.

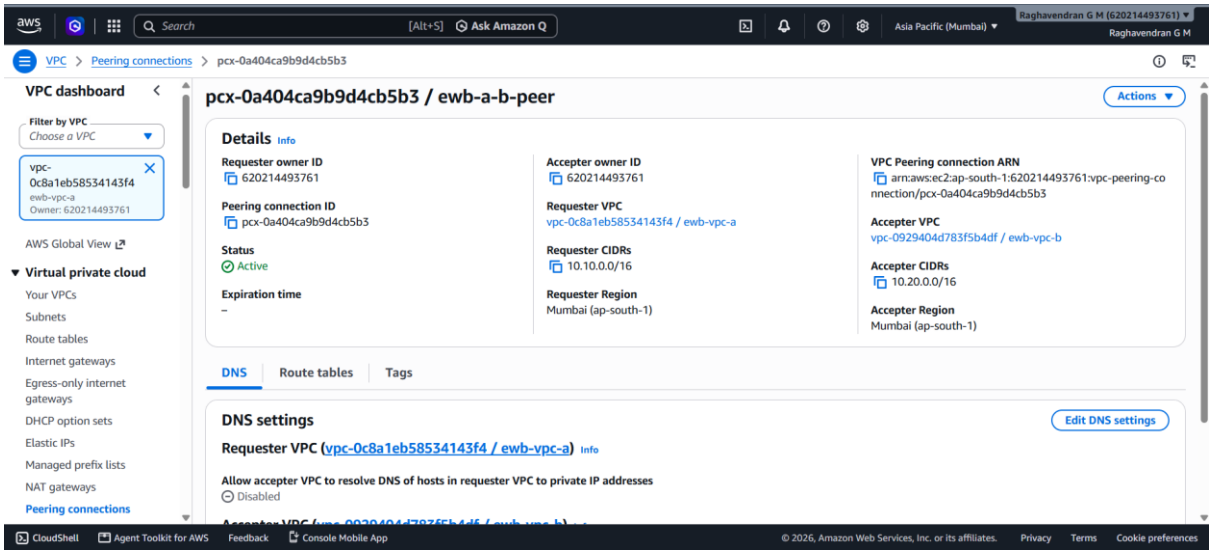
Forward Route Configuration (ewb-rt-a-private)

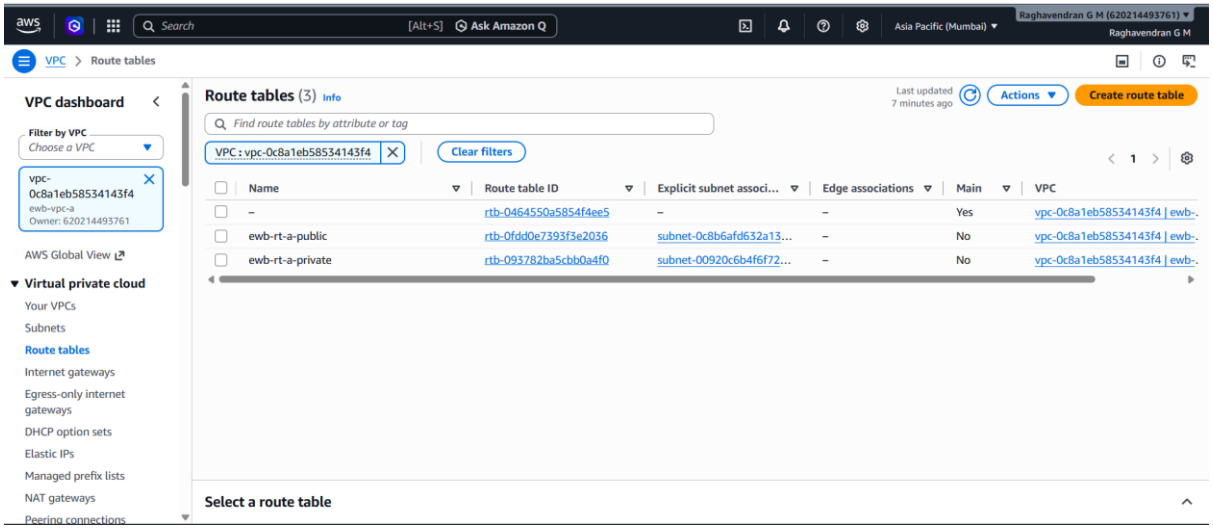
- **Route Table ID:** rtb-093782ba5cbb0a4f0
- **Target:** pcx-0a404ca9b9d4cb5b3
- **Destination:** 10.20.0.0/16 (Directs traffic intended for VPC B through the peering connection)

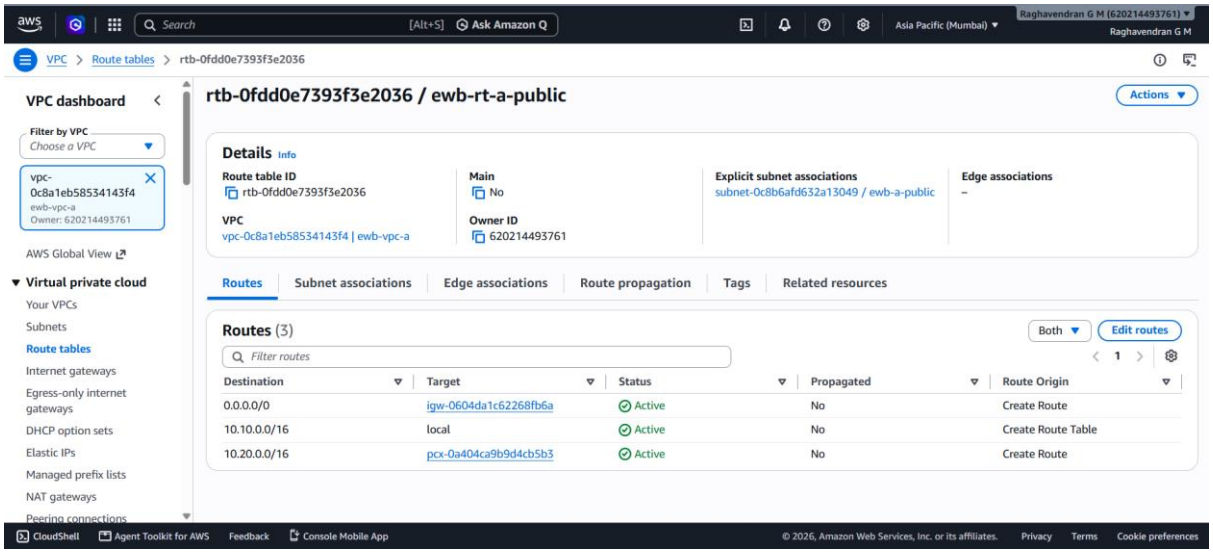
Return Route Configuration (ewb-rt-b-private)

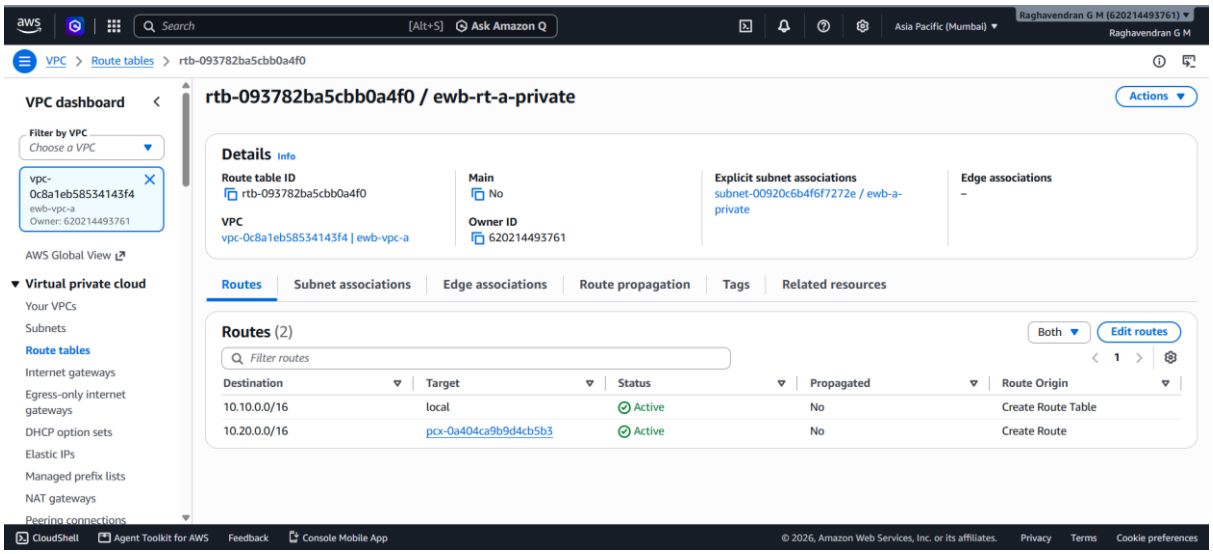
- **Route Table ID:** rtb-076064191b58b2
- **Target:** pcx-0a404ca9b9d4cb5b3
- **Destination:** 10.10.0.0/16 (Directs return traffic intended for VPC A back through the peering connection)

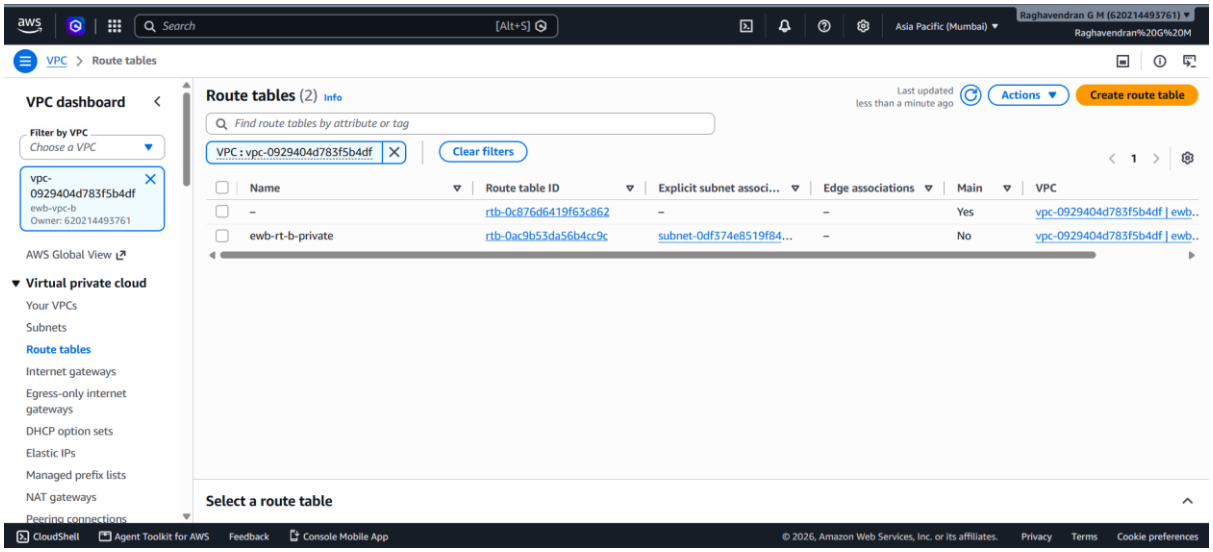

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pcx-0a404ca9b9d4cb5b3
Active
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620214493761
Peering connection ID
pcx-0a404ca9b9d4cb5b3
Status
Active
Expiration time
–
Accepter owner ID
620214493761
Requester VPC
vpc-0c8a1eb58534143f4 / ewb-vpc-a
Requester CIDRs
10.10.0.0/16
Requester Region
Mumbai (ap-south-1)
VPC Peering connection ARN
arn:aws:ec2:ap-south-1:620214493761:vpc-peering-connection/pcx-0a404ca9b9d4cb5b3
Accepter VPC
vpc-0929404d783f5b4df / ewb-vpc-b
Accepter CIDRs
10.20.0.0/16
Accepter Region
Mumbai (ap-south-1)
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Edit DNS settings
Requester VPC (vpc-0c8a1eb58534143f4 / ewb-vpc-a) Info
Allow accepter VPC to resolve DNS of hosts in requester VPC to private IP addresses
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Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
– rtb-0464550a5854f4ee5 – – Yes vpc-0c8a1eb58534143f4 | ewb-...
ewb-rt-a-public rtb-0fdd0e7393f3e2036 subnet-0c8b6afd632a13... – No vpc-0c8a1eb58534143f4 | ewb-...
ewb-rt-a-private rtb-093782ba5cbb0a4f0 subnet-00920c6b4f6f72... – No vpc-0c8a1eb58534143f4 | ewb-...
Select a route table

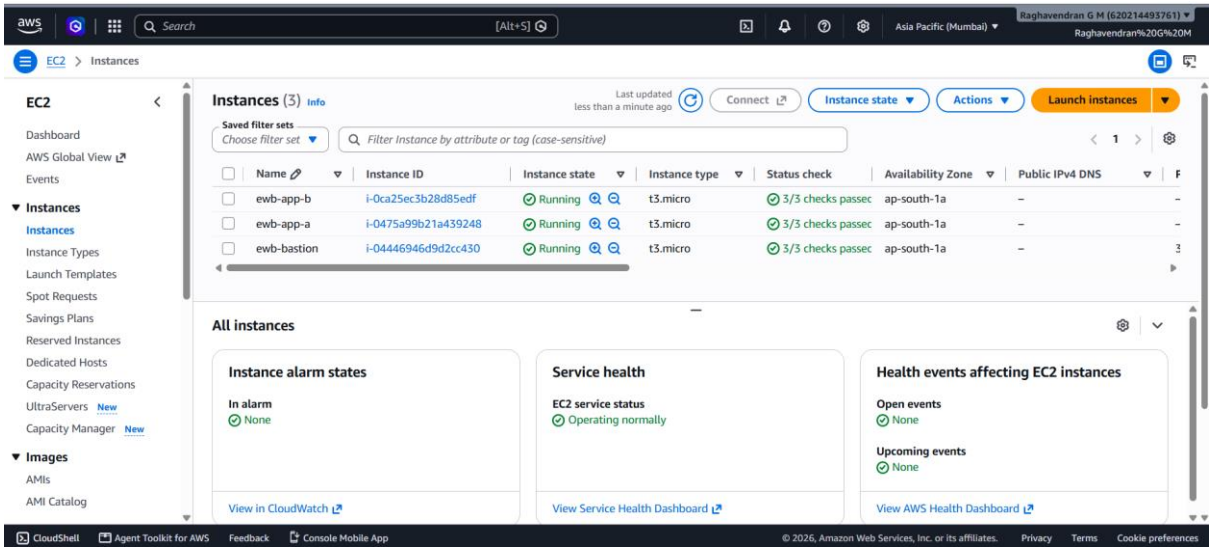

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Details Info
Route table ID
rtb-0fdd0e7393f3e2036
Main
No
Explicit subnet associations
subnet-0c8b6afd632a13049 / ewb-a-public
Edge associations
–
VPC
vpc-0c8a1eb58534143f4 | ewb-vpc-a
Owner ID
620214493761
Routes Subnet associations Edge associations Route propagation Tags Related resources
Routes (3)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0 igw-0604da1c62268fb6a Active No Create Route
10.10.0.0/16 local Active No Create Route Table
10.20.0.0/16 pcx-0a404ca9b9d4cb5b3 Active No Create Route
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Raghavendran G M
VPC > Route tables > rtb-093782ba5cbb0a4f0
VPC dashboard
rtb-093782ba5cbb0a4f0 / ewb-rt-a-private
Actions
Details Info
Route table ID
rtb-093782ba5cbb0a4f0
Main
No
Explicit subnet associations
subnet-00920c6b4f6f7272e / ewb-a-private
Edge associations
–
VPC
vpc-0c8a1eb58534143f4 | ewb-vpc-a
Owner ID
620214493761
Routes Subnet associations Edge associations Route propagation Tags Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
10.10.0.0/16 local Active No Create Route Table
10.20.0.0/16 pcx-0a404ca9b9d4cb5b3 Active No Create Route
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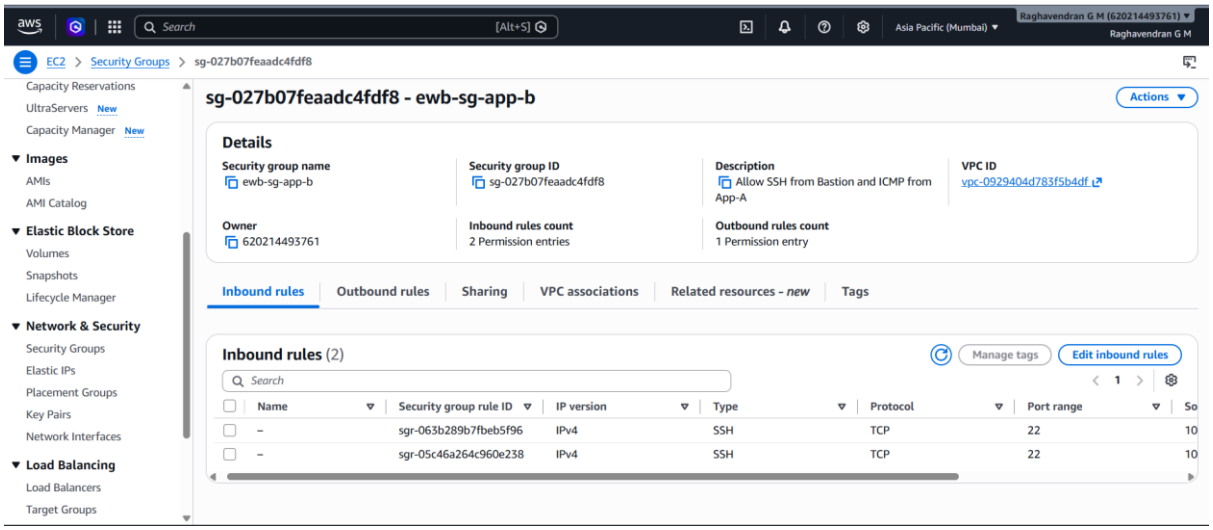
EC2 Instances & Security Groups

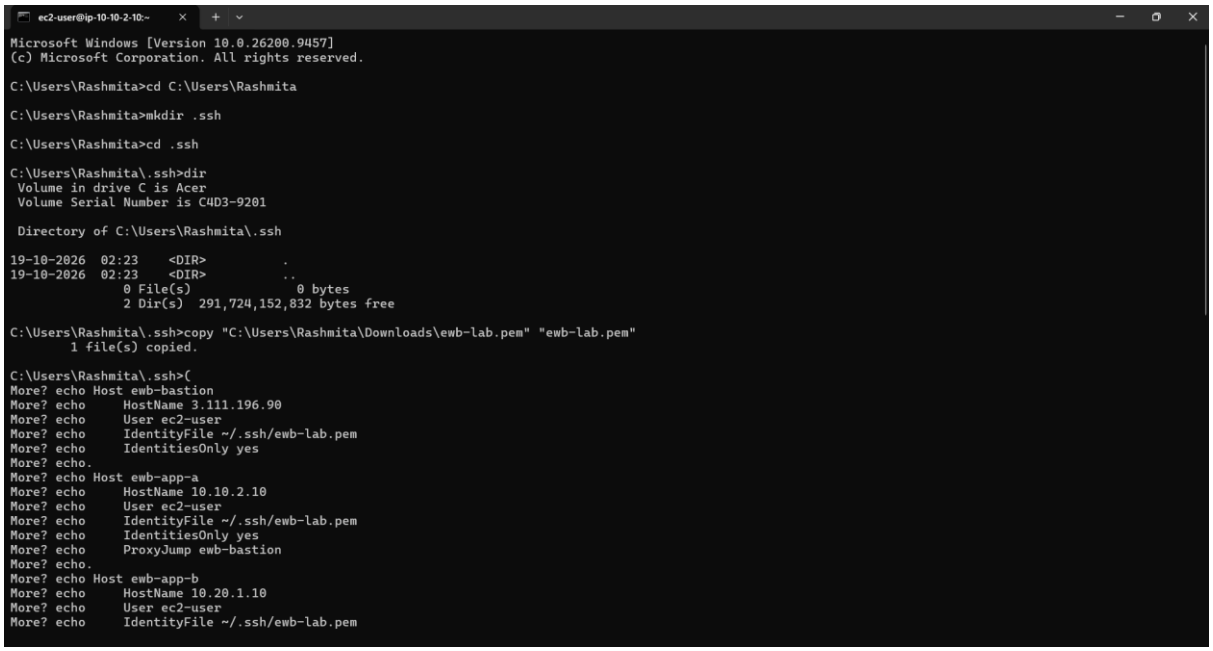
- **ewb-bastion (i-044063495f2a1c410):** Public IP 3.111.196.90 | Private IP 10.10.1.10
  - **SG (ewb-sg-bastion):** Inbound SSH (22) from 0.0.0.0/0
- **ewb-app-a (i-0475a8621a443524b):** Private IP 10.10.2.10
  - **SG (ewb-sg-app-a):** Inbound SSH (22) & ICMP from 10.10.1.10/32 & 10.20.0.0/16
- **ewb-app-b (i-0ca2fda18cf4b3b0d):** Private IP 10.20.1.10
  - **SG (ewb-sg-app-b):** sg-027b07feaadc4fdf8 | Inbound SSH (22) from 10.10.1.10/32 & 10.10.0.0/16, Custom ICMP from 10.10.0.0/16

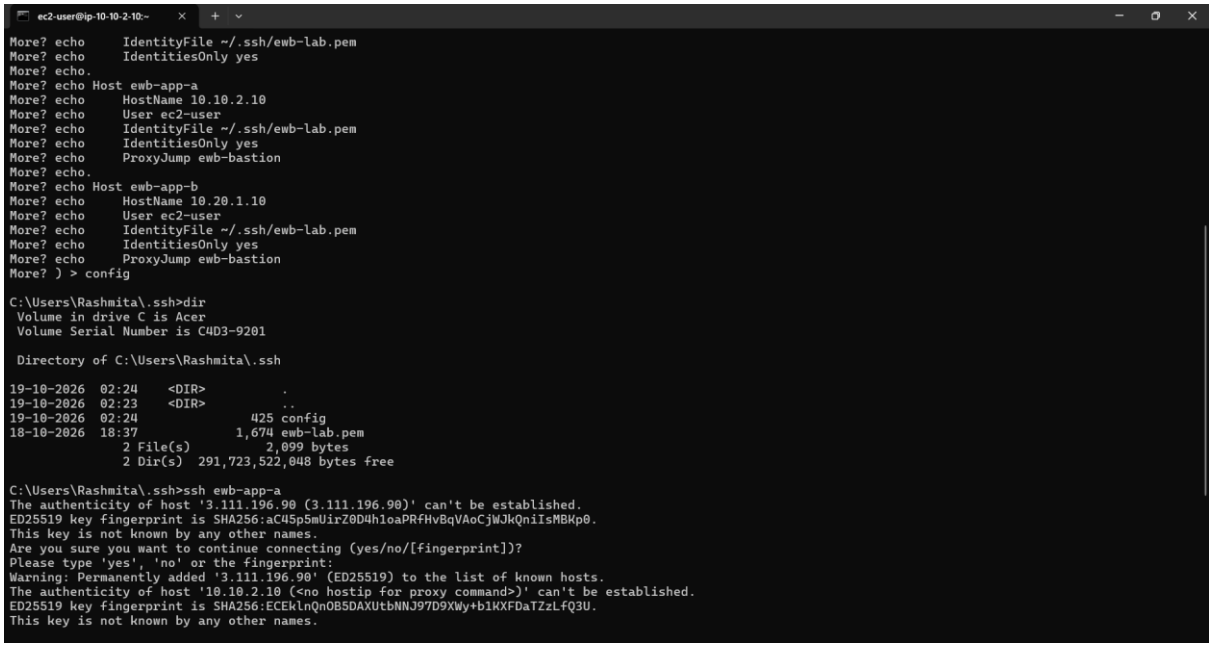


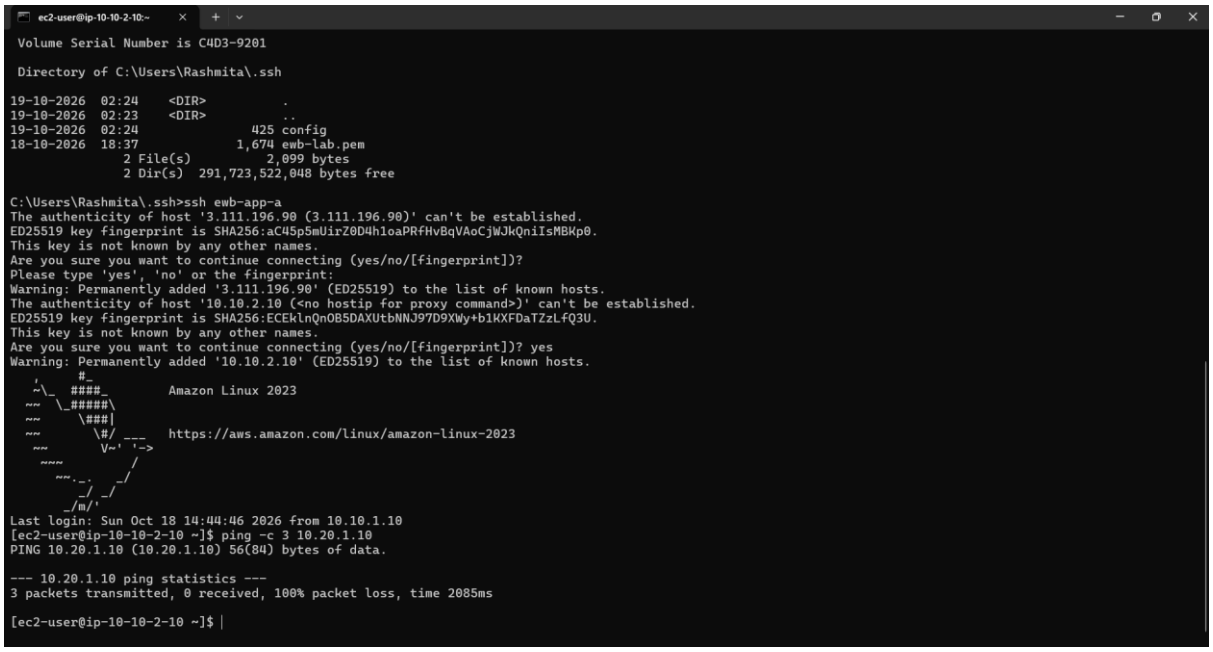
Controlled Failure & Recovery Analysis

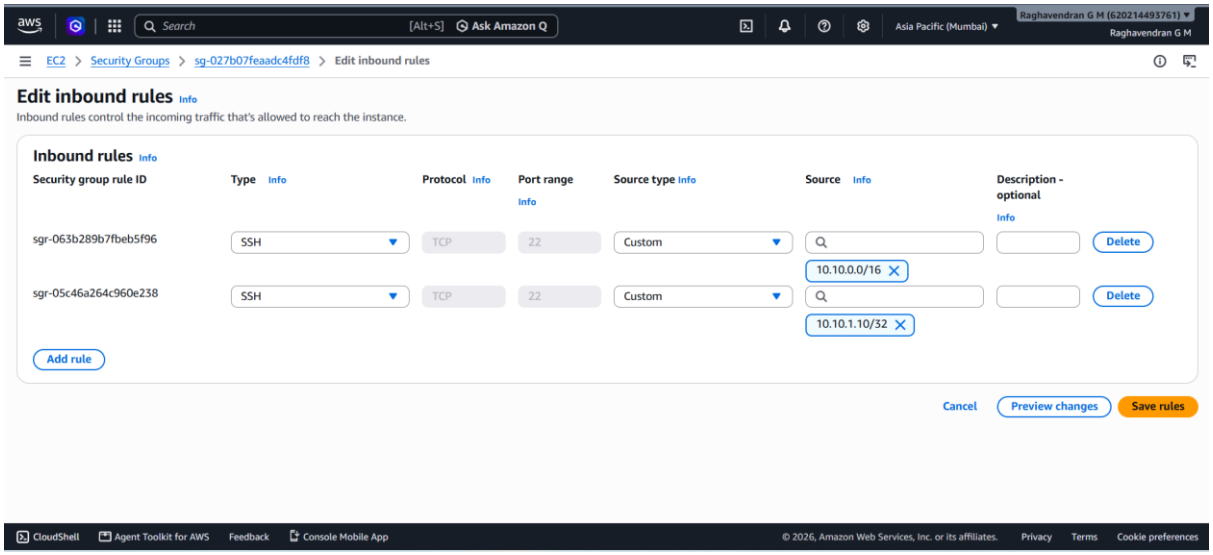
1. **Action:** Removed Custom ICMP - IPv4 rule from Security Group ewb-sg-app-b (sg-027b07feaadc4fdf8).
2. **Verification:** Executed ping -c 3 10.20.1.10 from ewb-app-a.
3. **Result:** 100% packet loss (3 transmitted, 0 received). Proves Security Groups act as stateful ENI firewalls dropping unauthorized packets regardless of active VPC routes.





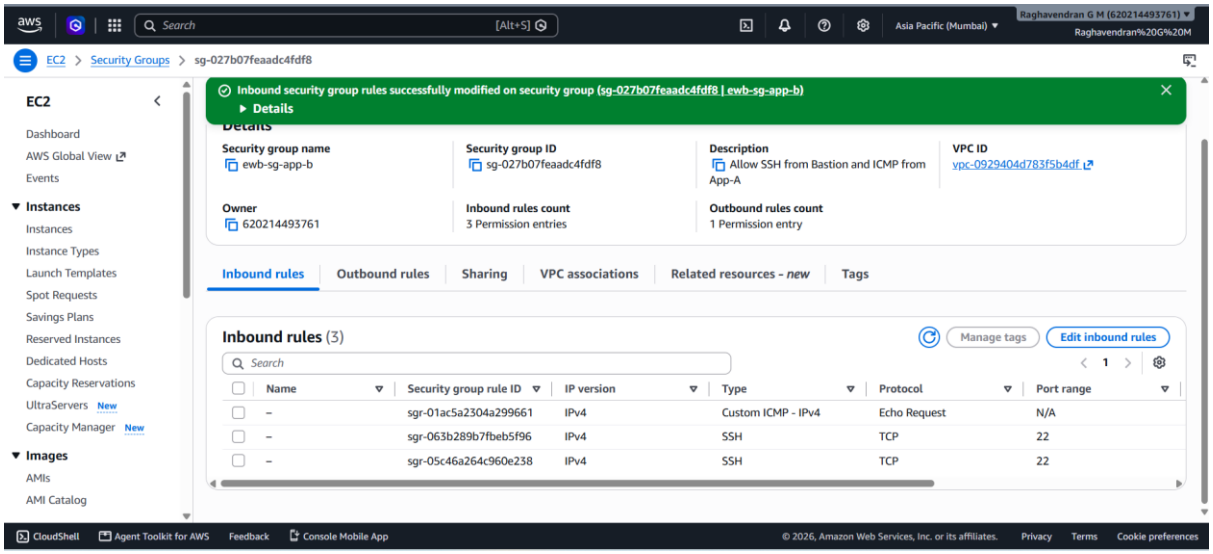

ec2-user@ip-10-10-2-10:~
More? echo IdentityFile ~/.ssh/ewb-lab.pem
More? echo IdentitiesOnly yes
More? echo.
More? echo Host ewb-app-a
More? echo HostName 10.10.2.10
More? echo User ec2-user
More? echo IdentityFile ~/.ssh/ewb-lab.pem
More? echo IdentitiesOnly yes
More? echo ProxyJump ewb-bastion
More? echo.
More? echo Host ewb-app-b
More? echo HostName 10.20.1.10
More? echo User ec2-user
More? echo IdentityFile ~/.ssh/ewb-lab.pem
More? echo IdentitiesOnly yes
More? echo ProxyJump ewb-bastion
More?) > config
C:\Users\Rashmita\.ssh>dir
Volume in drive C is Acer
Volume Serial Number is C4D3-9201
Directory of C:\Users\Rashmita\.ssh
19-10-2026 02:24 <DIR> .
19-10-2026 02:23 <DIR> ..
19-10-2026 02:24 425 config
18-10-2026 18:37 1,674 ewb-lab.pem
2 File(s) 2,099 bytes
2 Dir(s) 291,723,522,048 bytes free
C:\Users\Rashmita\.ssh>ssh ewb-app-a
The authenticity of host '3.111.196.90 (3.111.196.90)' can't be established.
ED25519 key fingerprint is SHA256:aC45p5mUirZ0D4h1oaPRfHvBqVAoCjWJkQniIsMBKp0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Please type 'yes', 'no' or the fingerprint:
Warning: Permanently added '3.111.196.90' (ED25519) to the list of known hosts.
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:ECEklnQnOB5DAXUtbNNJ97D9XWy+b1KXFDaTZzLfQ3U.
This key is not known by any other names.

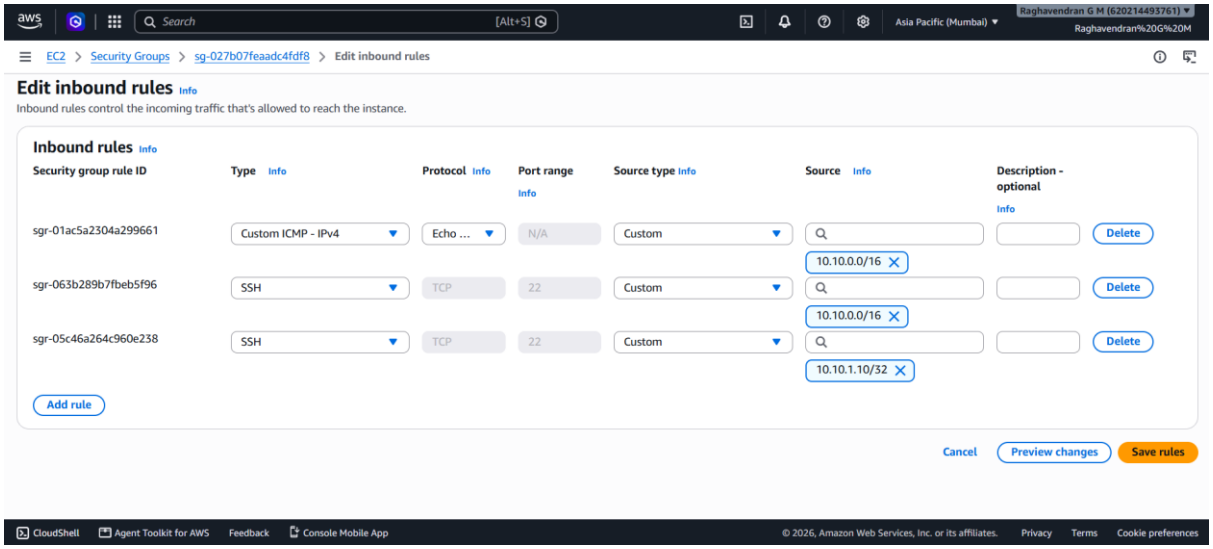

ec2-user@ip-10-10-2-10:~
Volume Serial Number is C4D3-9201
Directory of C:\Users\Rashmita\.ssh
19-10-2026 02:24 <DIR> .
19-10-2026 02:23 <DIR> ..
19-10-2026 02:24 425 config
18-10-2026 18:37 1,674 ewb-lab.pem
2 File(s) 2,099 bytes
2 Dir(s) 291,723,522,048 bytes free
C:\Users\Rashmita\.ssh>ssh ewb-app-a
The authenticity of host '3.111.196.90 (3.111.196.90)' can't be established.
ED25519 key fingerprint is SHA256:aC45p5mUirZ0D4h1oaPRfHvBqVAoCjWJkQniIsMBKp0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Please type 'yes', 'no' or the fingerprint:
Warning: Permanently added '3.111.196.90' (ED25519) to the list of known hosts.
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:ECEklnQnOB5DAXUtbNNJ97D9XWy+b1KXFDaTZzLfQ3U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sun Oct 18 14:44:46 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 3 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2085ms
[ec2-user@ip-10-10-2-10 ~]$

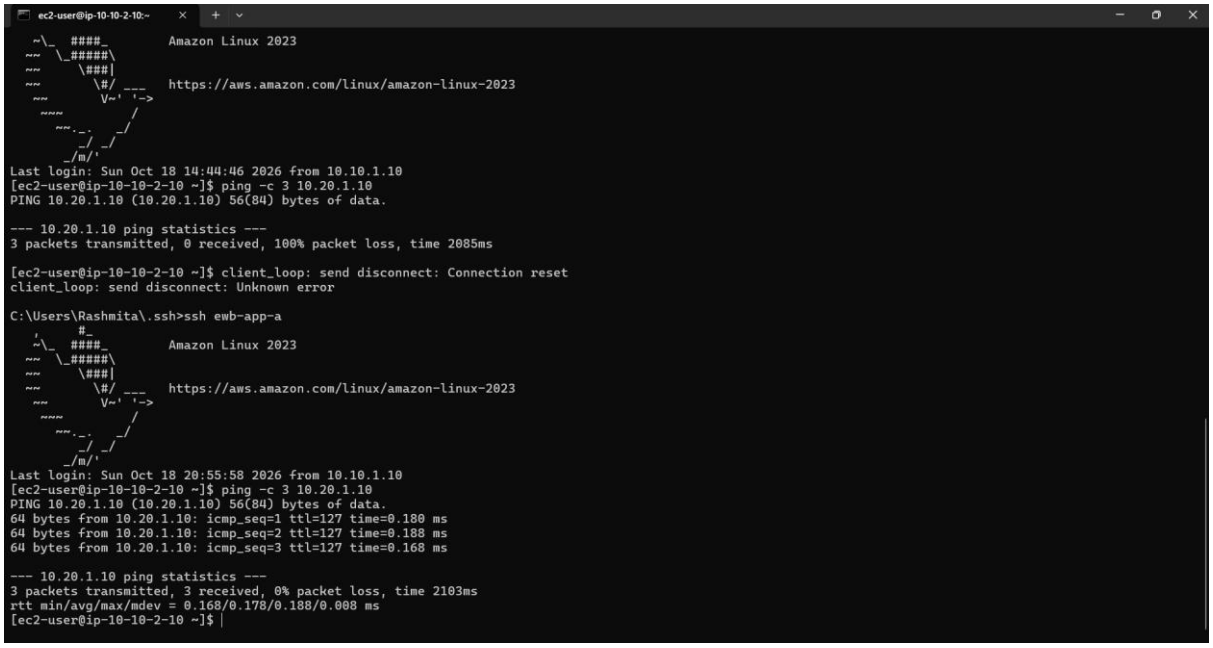


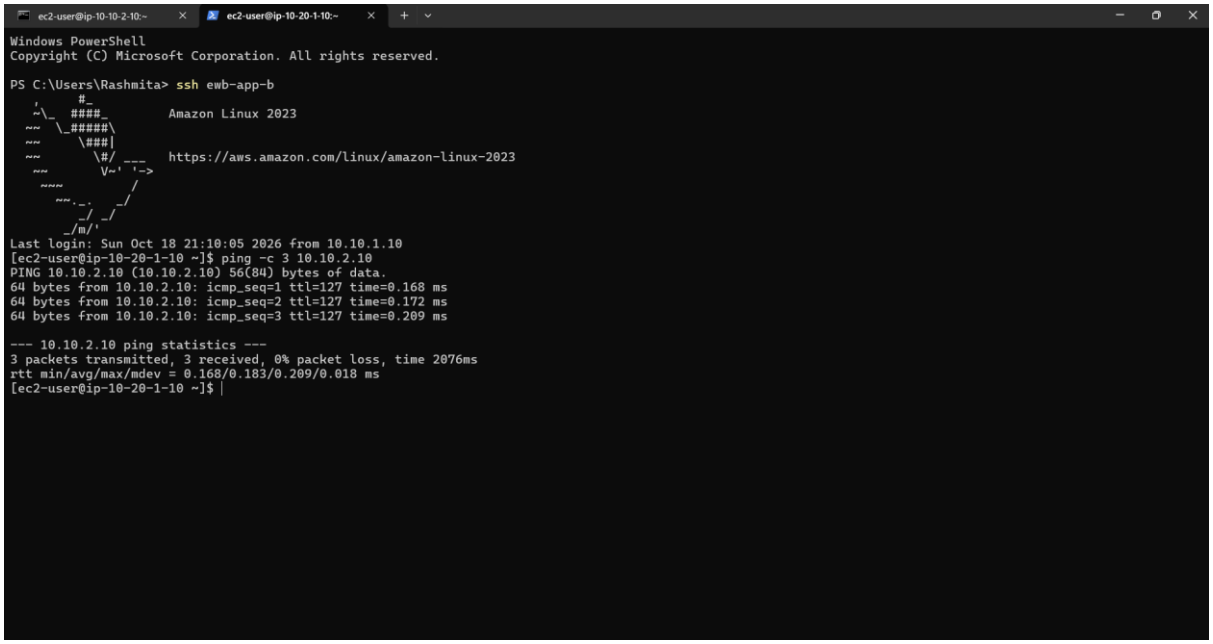
Recovery Verification

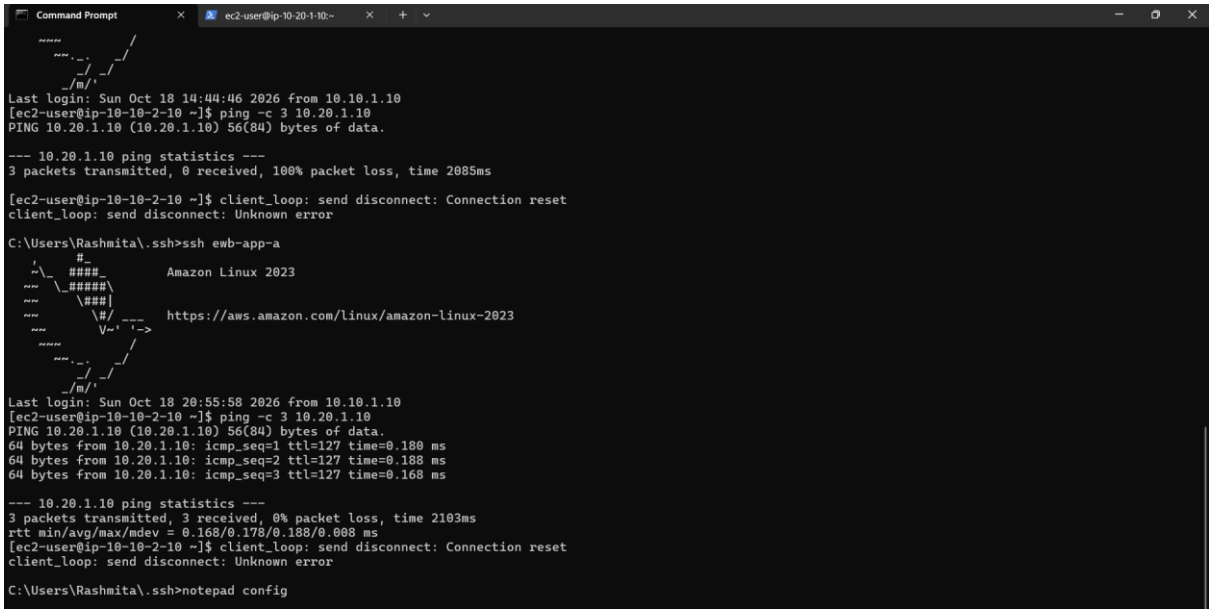
1. **Action:** Restored Custom ICMP - IPv4 rule with source 10.10.0.0/16 in ewb-sg-app-b.
2. **Verification:** Re-executed ping -c 3 10.20.1.10 from ewb-app-a.
3. **Result:** 0% packet loss (3 received, average rtt = 0.178 ms). Bidirectional reachability verified.

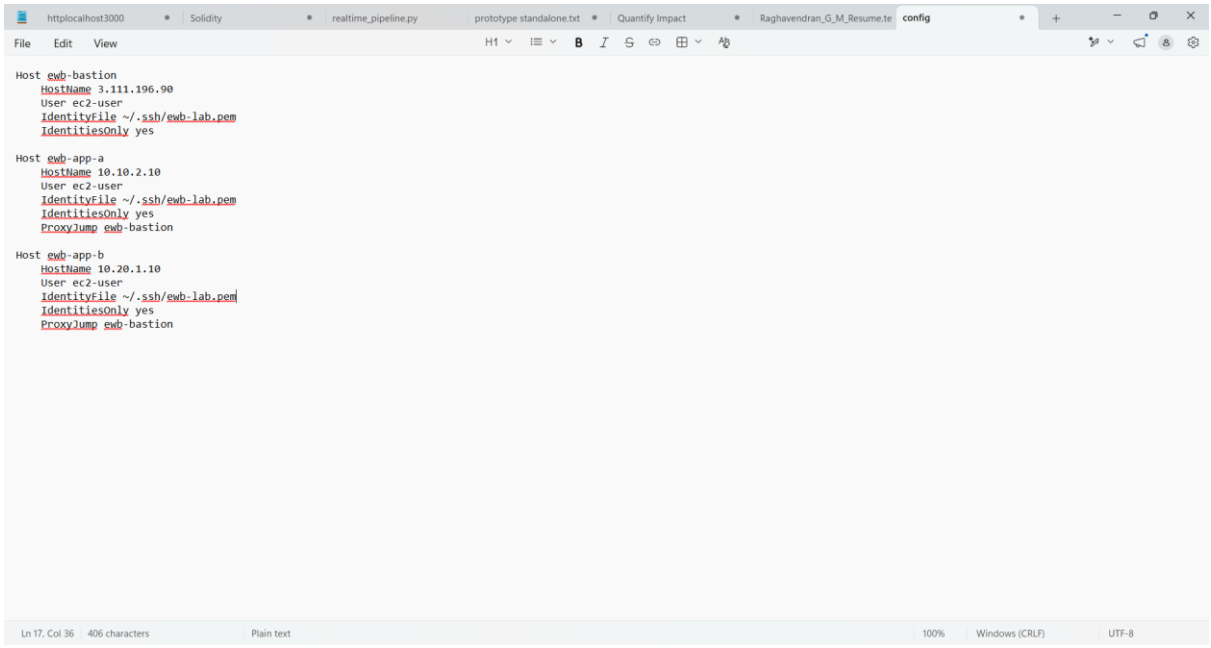



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EC2 > Security Groups > sg-027b07feaadc4fdf8 > Edit inbound rules
Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source type Info
Source Info
Description - optional Info
sgr-01ac5a2304a299661
Custom ICMP - IPv4
Echo ...
N/A
Custom
10.10.0.0/16
Delete
sgr-063b289b7fbeb5f96
SSH
TCP
22
Custom
10.10.0.0/16
Delete
sgr-05c46a264c960e238
SSH
TCP
22
Custom
10.10.1.10/32
Delete
Add rule
Cancel
Preview changes
Save rules
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ec2-user@ip-10-10-2-10:~
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sun Oct 18 14:44:46 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 3 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2085ms
[ec2-user@ip-10-10-2-10 ~]$ client_loop: send disconnect: Connection reset
client_loop: send disconnect: Unknown error
C:\Users\Rashmita\.ssh>ssh ewb-app-a
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sun Oct 18 20:55:58 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 3 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.180 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.188 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.168 ms
--- 10.20.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2103ms
rtt min/avg/max/mdev = 0.168/0.178/0.188/0.008 ms
[ec2-user@ip-10-10-2-10 ~]$


ec2-user@ip-10-10-2-10:~
ec2-user@ip-10-20-1-10:~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\Rashmita> ssh ewb-app-b
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sun Oct 18 21:10:05 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 3 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.168 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.172 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.209 ms
--- 10.10.2.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2076ms
rtt min/avg/max/mdev = 0.168/0.183/0.209/0.018 ms
[ec2-user@ip-10-20-1-10 ~]$


Command Prompt
ec2-user@ip-10-20-1-10:~
Last login: Sun Oct 18 14:44:46 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 3 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2085ms
[ec2-user@ip-10-10-2-10 ~]$ client_loop: send disconnect: Connection reset
client_loop: send disconnect: Unknown error
C:\Users\Rashmita\.ssh>ssh ewb-app-a
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sun Oct 18 20:55:58 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 3 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.180 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.188 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.168 ms
--- 10.20.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2103ms
rtt min/avg/max/mdev = 0.168/0.178/0.188/0.008 ms
[ec2-user@ip-10-10-2-10 ~]$ client_loop: send disconnect: Connection reset
client_loop: send disconnect: Unknown error
C:\Users\Rashmita\.ssh>notepad config



Resource Cleanup & Cloud FinOps Verification

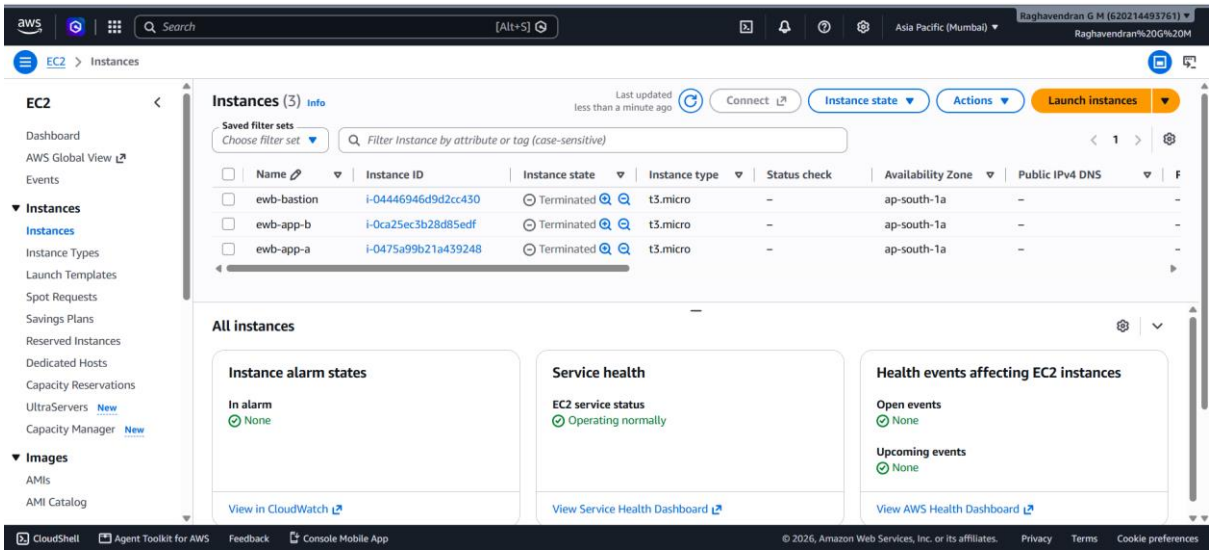
Teardown Steps

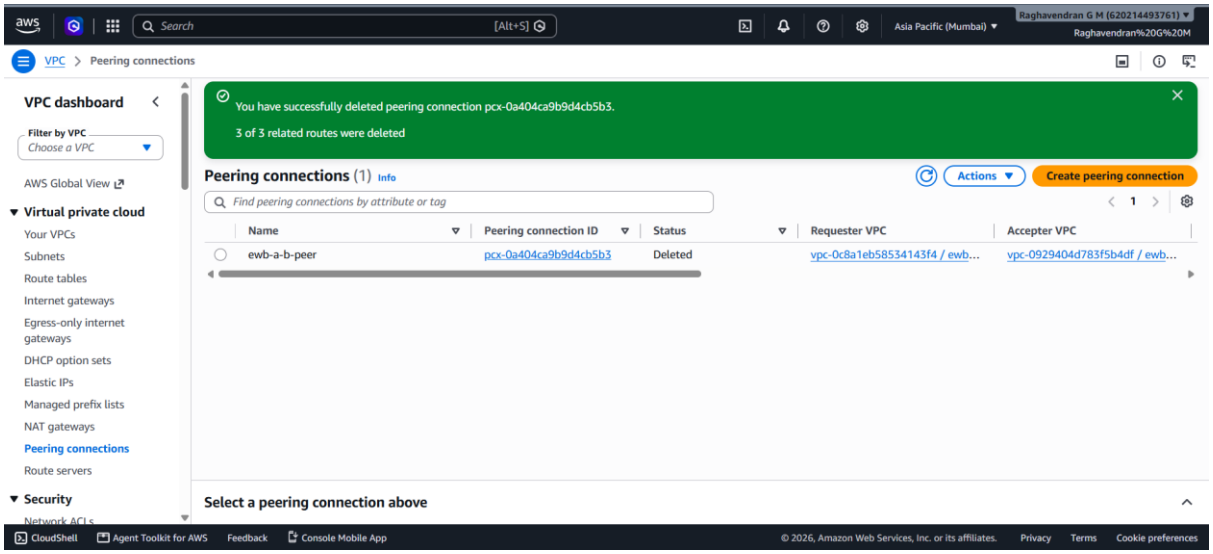
1.EC2 Instances: Terminated ewb-bastion, ewb-app-a, and ewb-app-b.

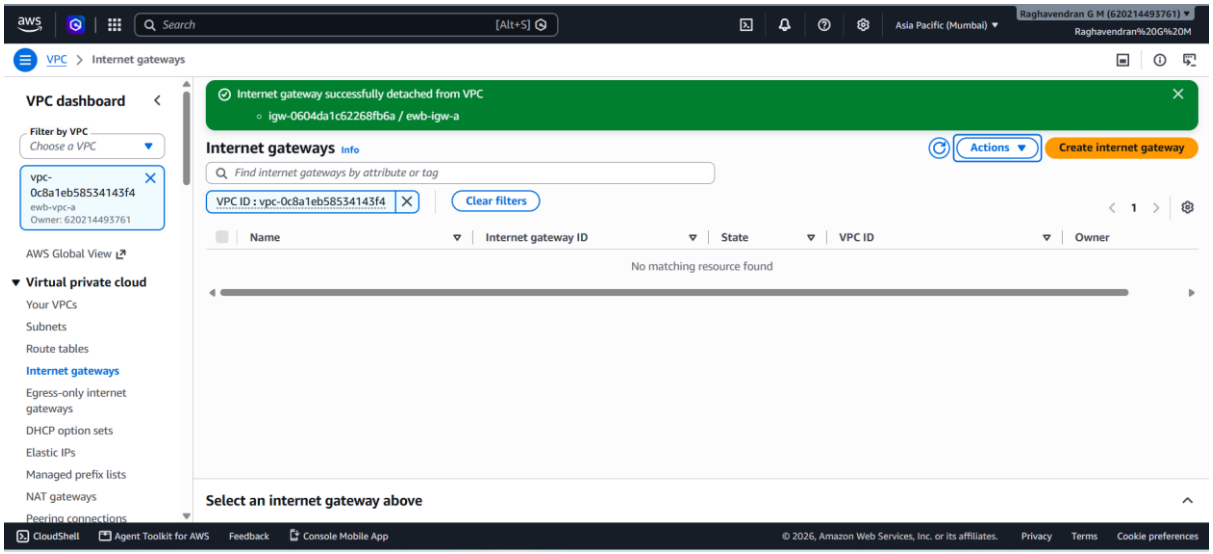
2.Peering Connection: Deleted pcx-0a404ca9b9d4cb5b3 (ewb-a-b-peer).

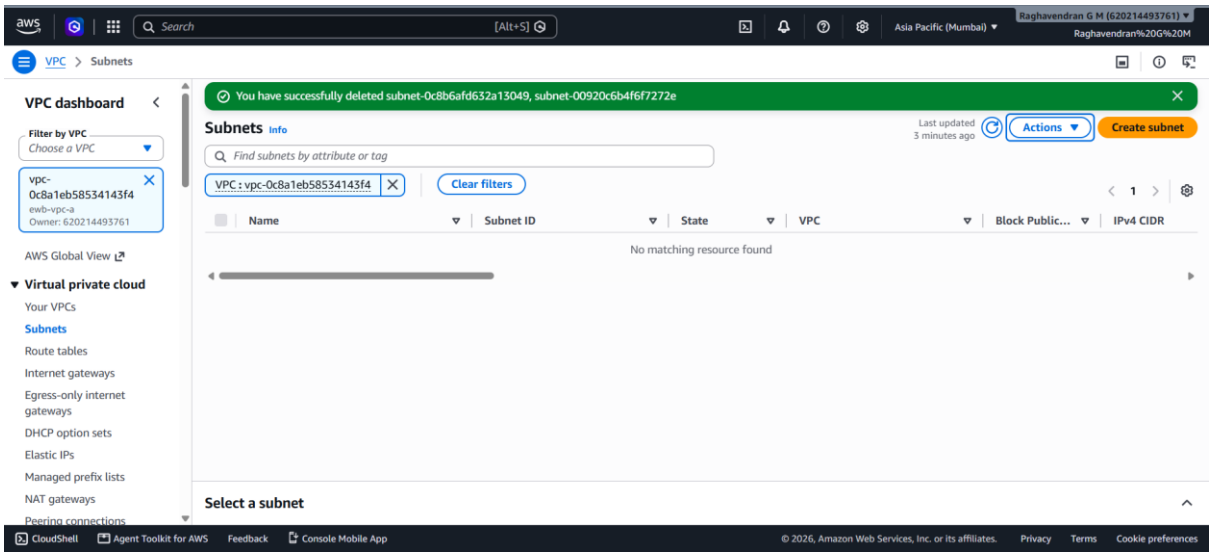
3.Gateways & Routing: Detached and deleted igw-0604da1c62268fb6a, custom route tables (ewb-rt-a-public, ewb-rt-a-private, ewb-rt-b-private), subnets, and Security Groups.

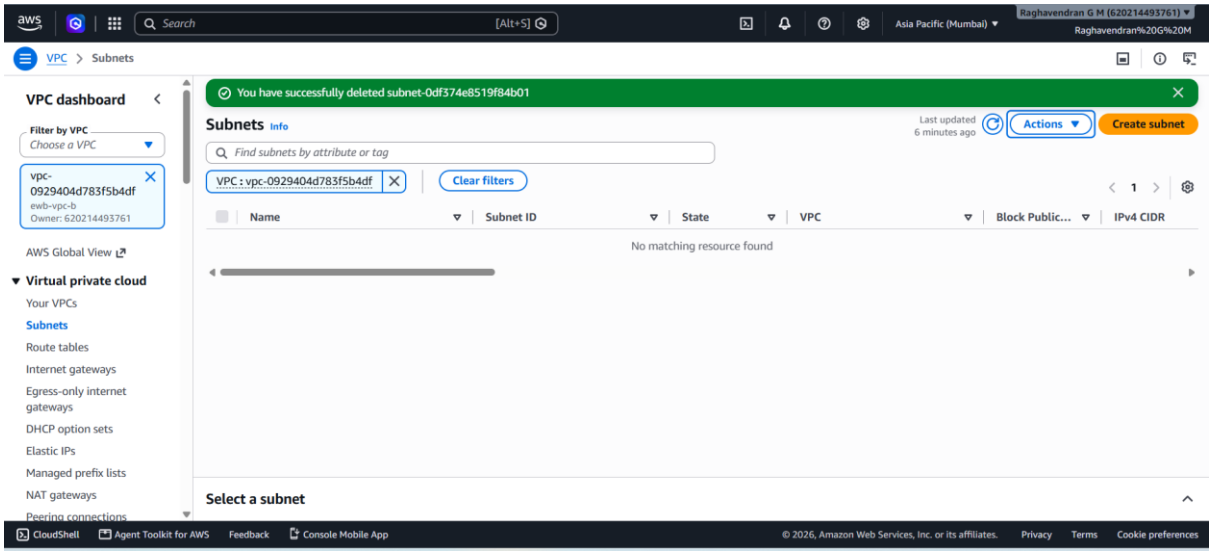
4.VPCs: Deleted ewb-vpc-a and ewb-vpc-b**.**

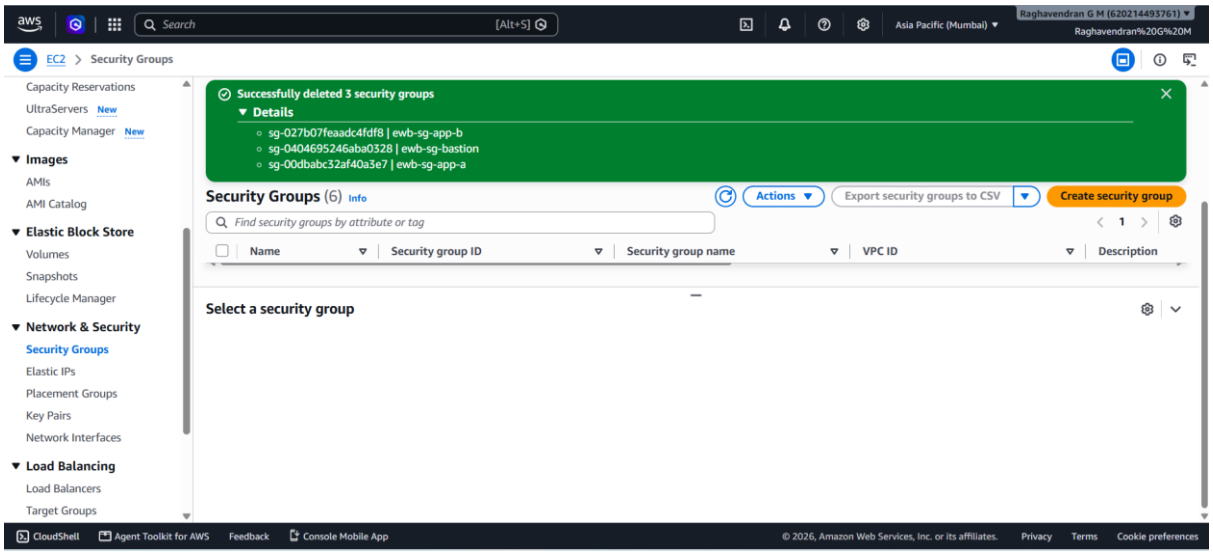


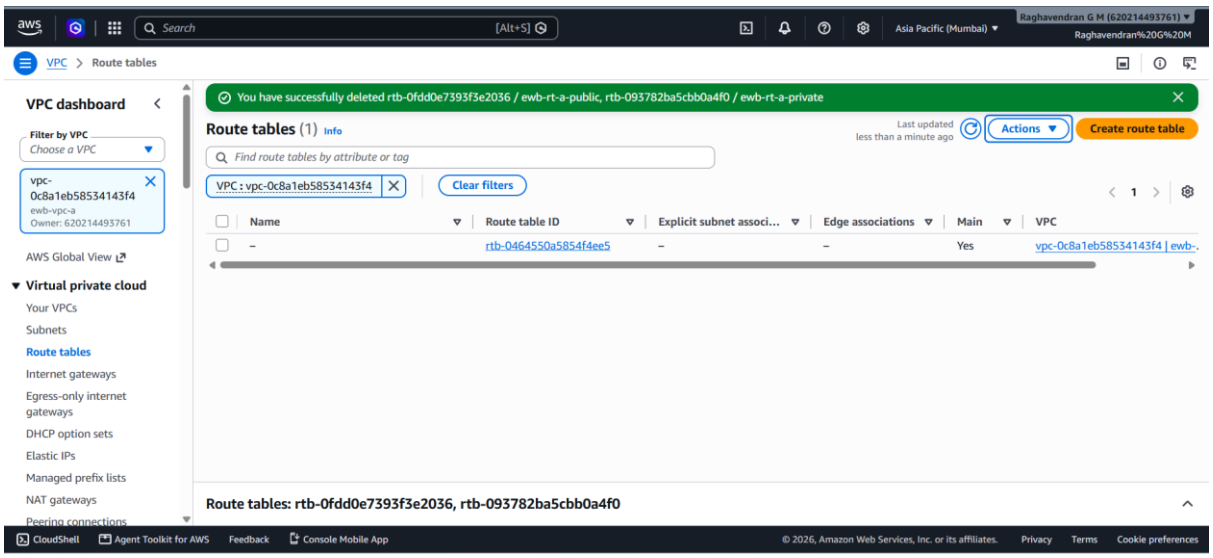

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VPC > Peering connections
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
You have successfully deleted peering connection pcx-0a404ca9b9d4cb5b3.
3 of 3 related routes were deleted
Peering connections (1) Info
Find peering connections by attribute or tag
Actions
Create peering connection
Name
Peering connection ID
Status
Requester VPC
Accepter VPC
ewb-a-b-peer
pcx-0a404ca9b9d4cb5b3
Deleted
vpc-0c8a1eb58534143f4 / ewb...
vpc-0929404d783f5b4df / ewb...
Select a peering connection above
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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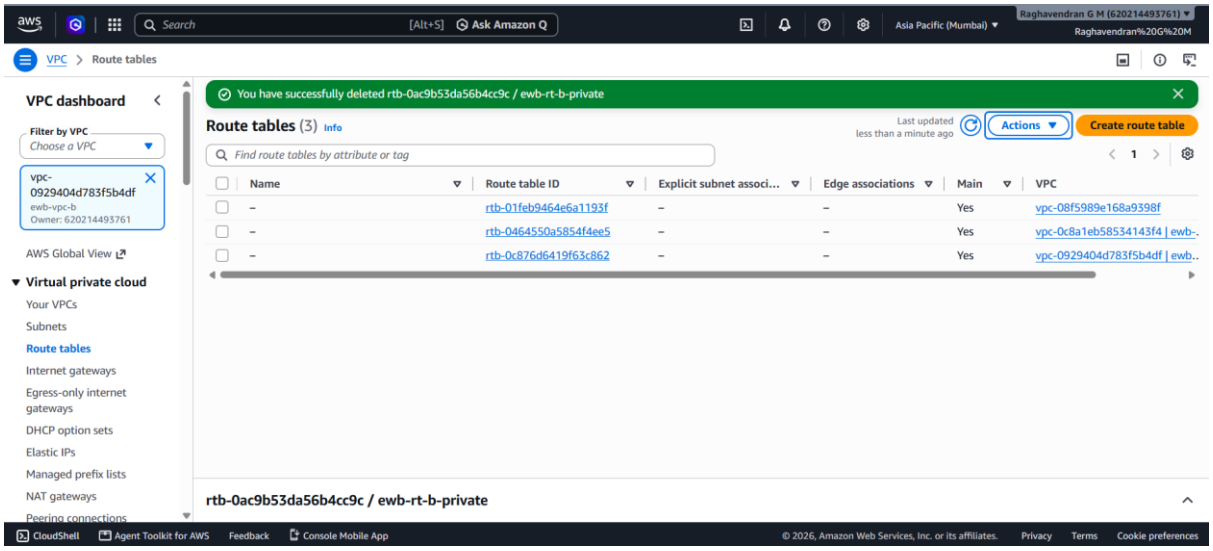

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VPC > Internet gateways
VPC dashboard
Filter by VPC
Choose a VPC
vpc-0c8a1eb58534143f4
ewb-vpc-a
Owner: 620214493761
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Internet gateway successfully detached from VPC
igw-0604da1c62268fb6a / ewb-igw-a
Internet gateways Info
Find internet gateways by attribute or tag
Actions
Create internet gateway
VPC ID : vpc-0c8a1eb58534143f4
Clear filters
Name
Internet gateway ID
State
VPC ID
Owner
No matching resource found
Select an internet gateway above
CloudShell
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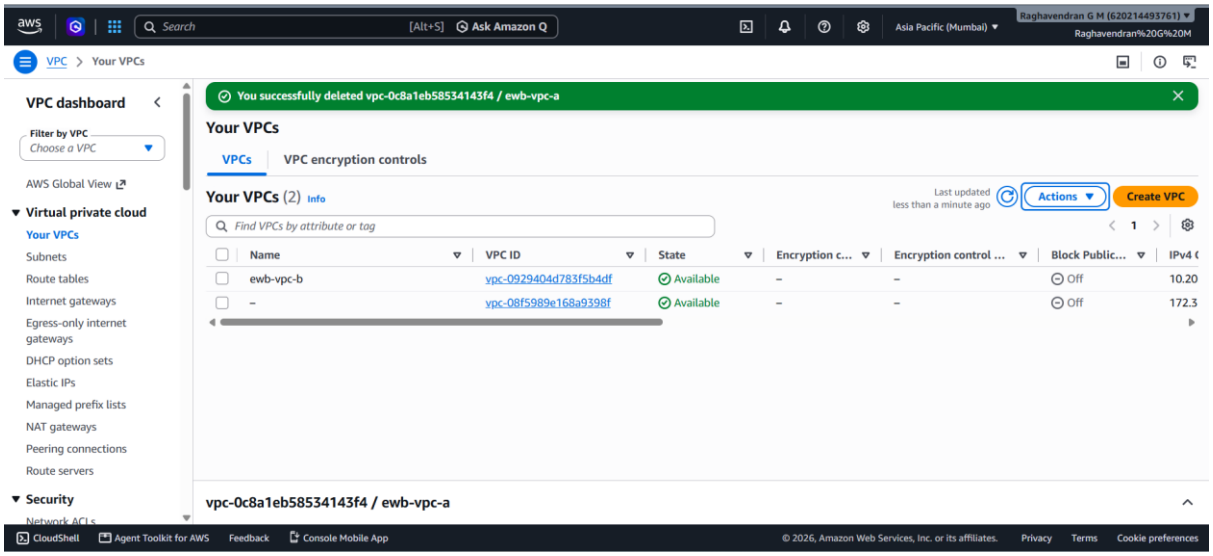

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VPC > Subnets
VPC dashboard
Filter by VPC
Choose a VPC
vpc-0c8a1eb58534143f4
ewb-vpc-a
Owner: 620214493761
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
You have successfully deleted subnet-0c8b6afd632a13049, subnet-00920c6b4f6f7272e
Subnets Info
Last updated 3 minutes ago
Actions
Create subnet
Find subnets by attribute or tag
VPC : vpc-0c8a1eb58534143f4
Clear filters
Name
Subnet ID
State
VPC
Block Public...
IPv4 CIDR
No matching resource found
Select a subnet
CloudShell
Agent Toolkit for AWS
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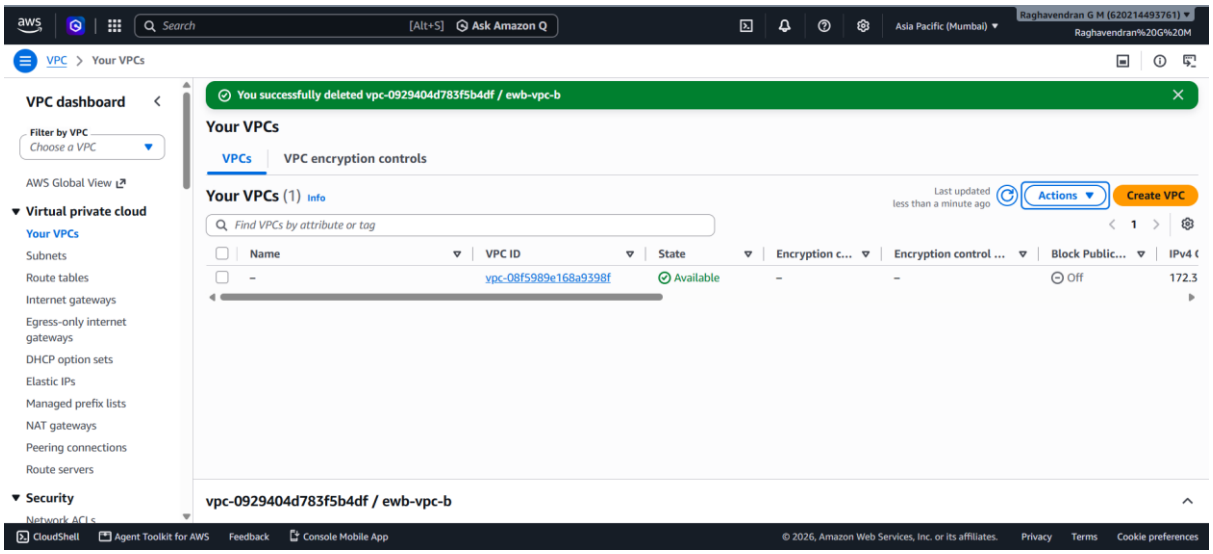

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VPC > Subnets
VPC dashboard
You have successfully deleted subnet-0df374e8519f84b01
Subnets Info
Last updated 6 minutes ago
Actions
Create subnet
Find subnets by attribute or tag
VPC : vpc-0929404d783f5b4df
Clear filters
Name
Subnet ID
State
VPC
Block Public...
IPv4 CIDR
No matching resource found
Select a subnet
Filter by VPC
Choose a VPC
vpc-0929404d783f5b4df
ewb-vpc-b
Owner: 620214493761
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EC2 > Security Groups
Successfully deleted 3 security groups
Details
sg-027b07feaadc4fdf8 | ewb-sg-app-b
sg-0404695246aba0328 | ewb-sg-bastion
sg-00dbabc32af40a3e7 | ewb-sg-app-a
Security Groups (6) Info
Actions
Export security groups to CSV
Create security group
Find security groups by attribute or tag
Name
Security group ID
Security group name
VPC ID
Description
Select a security group
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
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VPC > Route tables
VPC dashboard
You have successfully deleted rtb-0fdd0e7393f3e2036 / ewb-rt-a-public, rtb-093782ba5cbb0a4f0 / ewb-rt-a-private
Route tables (1) Info
Last updated less than a minute ago
Actions
Create route table
Find route tables by attribute or tag
VPC : vpc-0c8a1eb58534143f4
Clear filters
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
rtb-0464550a5854f4ee5
Yes
vpc-0c8a1eb58534143f4 | ewb-...
Route tables: rtb-0fdd0e7393f3e2036, rtb-093782ba5cbb0a4f0
Filter by VPC
Choose a VPC
vpc-0c8a1eb58534143f4
ewb-vpc-a
Owner: 620214493761
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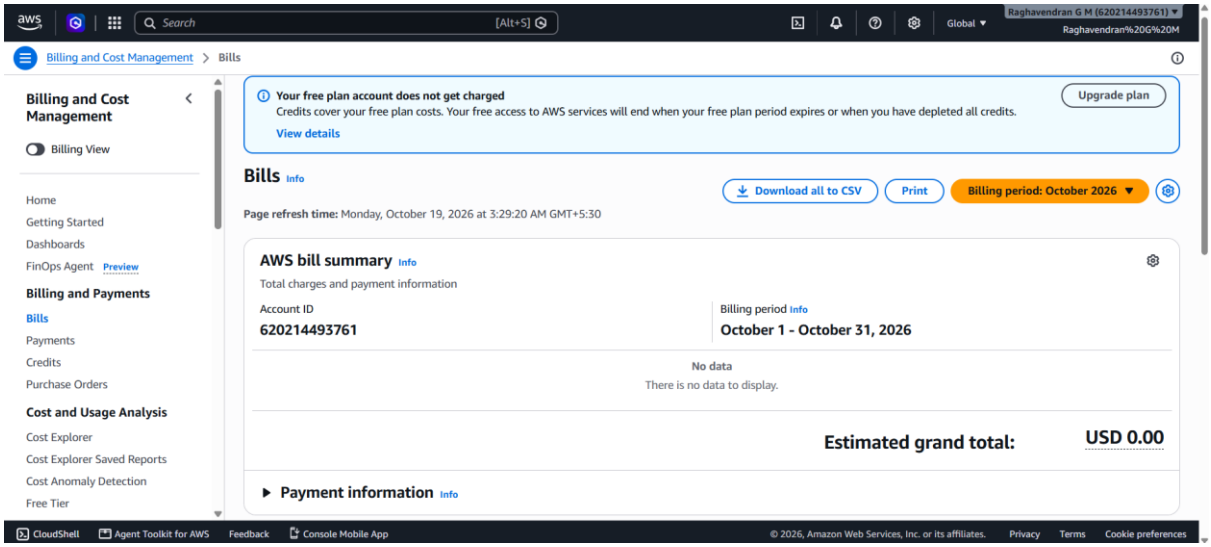

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VPC > Route tables
VPC dashboard
Filter by VPC
Choose a VPC
vpc-0929404d783f5b4df
ewb-vpc-b
Owner: 620214493761
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DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
You have successfully deleted rtb-0ac9b53da56b4cc9c / ewb-rt-b-private
Route tables (3) Info
Last updated less than a minute ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
rtb-01feb9464e6a1193f
rtb-0464550a5854f4ee5
rtb-0c876d6419f63c862
vpc-08f5989e168a9398f
vpc-0c8a1eb58534143f4 | ewb-..
vpc-0929404d783f5b4df | ewb..
rtb-0ac9b53da56b4cc9c / ewb-rt-b-private
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VPC > Your VPCs
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Peering connections
Route servers
Security
Network ACLs
You successfully deleted vpc-0c8a1eb58534143f4 / ewb-vpc-a
Your VPCs
VPCs
VPC encryption controls
Your VPCs (2) Info
Last updated less than a minute ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv4 (
ewb-vpc-b
vpc-0929404d783f5b4df
Available
Off
10.20
vpc-08f5989e168a9398f
172.3
vpc-0c8a1eb58534143f4 / ewb-vpc-a
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You successfully deleted vpc-0929404d783f5b4df / ewb-vpc-b
Your VPCs
VPCs
VPC encryption controls
Your VPCs (1) Info
Last updated less than a minute ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv4 (
vpc-08f5989e168a9398f
Available
Off
172.3
vpc-0929404d783f5b4df / ewb-vpc-b
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Billing Verification

- **Verified via AWS Billing and Cost Management > Bills:** Total active billable allocations returned to zero, resulting in $0.00 total charges under Free Tier coverage.



Key Engineering Takeaways

1. **Private AWS Backbone Traversal:** VPC Peering routes internal network traffic securely across the private AWS backbone without internet exposure.
2. **Symmetric Routing:** Bidirectional traffic requires matching destination routes in both requester and accepter route tables.
3. **Stateful Firewall Enforcement**: Security Groups enforce ENI-level filtration; missing rules block traffic even when VPC routes are active.
4. **Zero-Trust SSH Access:** Local SSH ProxyJump provides secure management of isolated instances without exposing SSH keys to intermediate public hosts.